SQL Notes

Converted from SEPT_SQL.txt

DATA

DATA IS A RAWFACT WHICH DESCRIBES ATTRIBUTE OF OBJECT/ENTITY.

ATTRIBUTE

 IT IS A PROPERTIES OF AN ENTITY.

ENTITY

ANYTHING WHICH IS PHYSICALLY PRESENT IS CALLED AS ENTITY/OBJECT.

DATABASE:

IT IS A PLACE WHERE WE STORE THE DATA IN SYSTEMATIC AND ORGANISED MANNER.

1.WHAT ARE THE BASIC OPERATIONS PERFORMED ON DATABASE??

  C-CREATE
  R-READ/RETRIEVE
  U-UPDATE
  D-DELETE

DBMS(DATABASE MANNAGEMENT SYSTEM)

>IT IS A SOFTWARE WHICH MAINTAIN AND MANNAGE THE DATABASE.

>THE FEATURES OF DBMS SOFTWARE ARE:

 i.SECURITY
 ii.AUTHORISATION

>THE DATA WILL BE STORED IN FILE FORMAT.

>WE CAN COMMUNICATE WITH DBMS SOFTWARE BY USING QUERY LANGUAGE..

RDBMS(RELATIONAL DATABASE MANNAGEMENT SYSTEM)

IT IS A DBMS SOFTWARE WHERE WE STORE DATA IN THE FORM OF TABLE/RELATION

>IF DBMS FOLLOWS RELATIONAL MODEL OR EF CODD RULE THEN WE CONSIDER IT AS RDBMS SOFTWARE

>HERE,WE CAN STORE THE DATA IN THE FORM OF TABLE/RELATION

>TO COMMUNICATE WITH RDBMS SOFTWARE WE USE SQL(STRUCTURED QUERY LANGUAGE).

E.F CODD RULE

1.DATA ENTERED INTO A CELL MUST BE SINGLE VALUE DATA OR ATOMIC DATA.
2.WE CAN ENTER THE DATA INTO MULTPLE TABLE ALSO WE CAN BUILD CONNECTION BETWEEN
  MULTIPLE TABLES BY USING KEY ATTRIBUTES.
3.WE CAN VALIDATE THE DATA BY USING DATATYPE AND CONSTRAINT.
4.FOR A TABLE DATATYPE IS MANDATORY CONSTRAINT IS OPTIONAL.

CHAR

>HERE WE CAN STORE UPTO 255 CHARACTERS.
>IN CASE OF CHAR DATATYPE MEMORY CANNOT BE GIVEN BACK TO MEMORY LOCATION FOR RE-USE
 PURPOSE.
>HENCE THERE IS A WASTAGE OF MEMORY.
>IT IS ALSO KNOWN AS FIXED LENGTH MEMORY.

VARCHAR

>HERE WE CAN STORE UPTO 65535 CHARACTERS.
>IN CASE OF VARCHAR DATATYPE MEMORY CAN BE GIVEN BACK TO MEMORY LOCATION FOR RE-USE PURPOSE.
>HENCE THERE IS NO WASTAGE OF MEMORY.
>IT IS ALSO KNOWN AS VARIABLE LENGTH MEMORY.

WRITE A DIFFRENCE BETWEEN CHAR AND VARCHAR DATATYPE??

UNIQUE

TO AVOID DUPLICATE VALUES WHICH ARE ENTERING INTO A COLUMN IN ORIGINAL TABLE WE USE UNIQUE CONSTRAINT.

NULL

>IT IS A KEYWORD WHICH REPRESENTS EMPTY SET
>IT DOESN'T OCCUPY ANY MEMORY INSIDE MEMORY LOCATION

>ANY OPERATION PERFORMED WITH NULL,THE RESULTANT WILL BE NULL.
>IN RDBMS TWO NULL' S ARE NOT SAME.

NOT NULL

TO AVOID NULL VALUES WHICH ARE ENTERING INTO A COLUMN IN ORIGINAL TABLE WE USE NOT NULL CONSTRAINT.

CHECK

TO PROVIDE USER DEFINED CONDITION FOR A COLUMN BASED ON OUR REQUIREMENT WE USE CHECK CONSTRAINT.

SYNTAX:

CHECK(CONDITION);

PRIMARY KEY

IT IS USED TO UNIQUELY IDENTIFY THE RECORDS FROM THE TABLE.

CHARACTERISTINCS OF PRIMARY KEY

>TO REPRESENT THE TABLE AMONG THE DATABASE WE NEED PRIMARY KEY
>IT IS A COMBINATION OF UNIQUE AND NOT NULL CONSTRAINT.
>A Table Can Have Only 1 Primary Key.
>If Two Columns Are Eligible To Become Primary Key Means We can Choose any one based on our
requirement.
>For a table Primary Key is not mandatory but Designwise preferrable.

FOREIGN KEY

To establish the connection between multiple tables we use FOREIGN KEY.

CHARACTERISTICS OF FOREIGN KEY

>A table can have N number of FOREIGN KEY.
>A Primary Key of a table can eligible to become FOREIGN KEY in another table.
>It may allow duplicate or null values.
>For a table FOREIGN KEY is not mandatory,but Designwise preferrable.
>It present in CHILD table but belongs to PARENT table.
>It also known as REFERENTIAL INTIGRITY CONSTRAINT.

SQL STATEMENTS:

1.DDL(DATA DEFINATION LANGUAGE)

1.CREATE
2.ALTER
3.TRUNCATE
4.DROP

2.DML(DATA MANIPULATION LANGUAGE)

1.INSERT
2.UPDATE
3.DELETE

3.TCL(TRANSACTION CONTROL LANGUAGE)

1.COMMIT
2.ROLLBACK
3.SAVEPOINT

4.DCL(DATA CONTROL LANGUAGE)

1.GRANT
2.REVOKE

5.DQL(DATA QUERY LANGUAGE)

1.SELECT
2.PROJECTION
3.SELECTION
4.JOINS

DDL(DATA DEFINATION LANGUAGE)

1.CREATE:

IT IS USED TO CREATE DATABASE AND ITS OBJECTS SUCH AS TABLE,VIEW,PROCEDURE,TRIGGER,FUNCTION AND USER IN
MYSQL.

TO CREATE DATABASE IN MYSQL:

SYNTAX:

```
CREATE DATABASE DATABASE_NAME;
```

DEMO:

```
CREATE DATABASE DEMO;
```

TO VIEW ALL THE DATABASES PRESENT IN MYSQL:

SYNTAX:

```
SHOW DATABASES;
```

TO ACCESS PARTICULAR DATABASE IN MYSQL:

SYNTAX:

```
USE DATABASE_NAME;
```

```
USE DEMO;
```

TO CLEAR SCREEN IN MYSQL:

SYNTAX:

```
SYSTEM CLS
```

TO CLOSE THE MYSQL:

SYNTAX:

```
EXIT
```

TO VIEW ALL THE TABLES PRESENT IN ONE DATABASE:

SYNTAX:

```
SHOW TABLES;
```

TO CREATE TABLE:

SYNTAX:1

```
CREATE TABLE TABLE_NAME
(
COLUMN_NAME_1 DATATYPE SIGNED/UNSIGNED NULL/NOT NULL,
COLUMN_NAME_2 DATATYPE SIGNED/UNSIGNED NULL/NOT NULL,
'
'
'
'
COLUMN_NAME_N DATATYPE SIGNED/UNSIGNED NULL/NOT NULL,
CONSTRAINT CONSTRAINT_NAME PRIMARY KEY(COLUMN_NAME),
CONSTRAINT CONSTRAINT_NAME UNIQUE(COLUMN_NAME),
CONSTRAINT CONSTRAINT_NAME CHECK(CONDITION),
CONSTRAINT FOREIGN KEY(COLUMN_NAME)REFERENCES PARENT_TABLE_NAME(COLUMN_NAME)
);
```

```
CREATE TABLE STUDENT
(
ID INT NOT NULL,
NAME VARCHAR(20) NOT NULL,
PHONE BIGINT NOT NULL,
CONSTRAINT P_ID PRIMARY KEY(ID),
CONSTRAINT U_PH UNIQUE(PHONE),
CONSTRAINT C_PH CHECK(LENGTH(PHONE)=10)
);
```

```
BRANCH:
-------
BID     INT         UNSIGNED NN  PRIMARY KEY----P_BID
BNAME   VARCHAR(20)          NN
ADDRESS LONGTEXT             NN
PINCODE INT         UNSIGNED NN  CHECK(LENGTH(PINCODE)=6),UNIQUE
                                 C_PIN                   U_PIN
```

```
CREATE TABLE BRANCH
(
BID INT UNSIGNED NOT NULL,
BNAME VARCHAR(20) NOT NULL,
ADDRESS LONGTEXT NOT NULL,
PINCODE INT UNSIGNED NOT NULL,
CONSTRAINT P_BID PRIMARY KEY(BID),
```

```
CONSTRAINT U_PIN UNIQUE(PINCODE),
CONSTRAINT C_PIN CHECK(LENGTH(PINCODE)=6)
);
```

```
PRODUCT:
--------
PID      INT          NN   PRIAMRY KEY-----P_PID
PNAME    VARCHAR(20)  NN
PRICE    DECIMAL(5,2) NN   CHECK(PEICE>0)------C_PRICE
QUANTITY INT          NULL CHECK(QUANTITY>0)---- C_QUANTITY
```

```
CREATE TABLE PRODUCT
(
PID INT NOT NULL,
PNAME VARCHAR(20) NOT NULL,
PRICE DECIMAL(5,2) NOT NULL,
QUANTITY INT NULL,
CONSTRAINT P_PID PRIMARY KEY(PID),
CONSTRAINT C_PRICE CHECK(PRICE>0),
CONSTRAINT C_QUANTITY CHECK(QUANTITY>0)
);
```

```
TO VIEW TABLE STRUCTURE:
------------------------
SYNTAX:
```

```
DESC TABLE_NAME;
```

```
BRANCH:
```

```
DESC BRANCH;
```

```
TO VIEW ALL THE CONSTRAINT_NAME OF THE CONSTRAINTS:
---------------------------------------------------
STEP:1
```

```
USE INFORMATION_SCHEMA;
```

```
STEP:2
```

```
SELECT * FROM TABLE_CONSTRAINTS;
```

```
SYNTAX:2 ---- TO CREATE TABLE
-----------------------------
```

```
CREATE TABLE TABLE_NAME
(
COLUMN_NAME_1 DATATYPE SIGNED/UNSIGNED NULL/NOT NULL CONSTRAINT,
COLUMN_NAME_2 DATATYPE SIGNED/UNSIGNED NULL/NOT NULL CONSTRAINT,
'
'
'
'
'
'
COLUMN_NAME_N DATATYPE SIGNED/UNSIGNED NULL/NOT NULL CONSTRAINT,
CONSTRAINT FOREIGN KEY(COLUMN_NAME)REFERENCES PARENT_TABLE_NAME(COLUMN_NAME)
);
```

```
FACULTY:
--------
FID    INT          NN PRIMARY KEY
FNAME  VARCHAR(20)  NN
PHONE  BIGINT       NN UNIQUE,CHECK(LENGTH(PHONE)=10)
```

```
CUSTOMER:
---------
CID      INT           NN   PRIMARY KEY AUTO_INCREMENT
CNAME    VARCHAR(20)   NN
PHONE    BIGINT        NN   UNIQUE CHECK(LENGTH(PHONE)=10)
ADDRESS  LONGTEXT      NN
EMAIL    VARCHAR(20)   NULL UNIQUE
BALANCE  DECIMAL(5,2)  NN   DEFAULT  '420'
GENDER   ENUM('M','F') NN
ID       INT           NN   FOREIGN KEY
```

```
CREATE TABLE CUSTOMER
(
CID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
CNAME VARCHAR(20) NOT NULL,
PHONE BIGINT NOT NULL UNIQUE CHECK(LENGTH(PHONE)=10),
ADDRESS LONGTEXT NOT NULL,
EMAIL VARCHAR(20) UNIQUE,
```

```
BALANCE DECIMAL(5,2) NOT NULL DEFAULT '420',
GENDER ENUM('M','F') NOT NULL,
ID INT NOT NULL,
CONSTRAINT FOREIGN KEY(ID)REFERENCES STUDENT(ID)
);
```

```
BANK:
-----
ACCOUNT_NUMBER       INT          NN PRIMARY KEY
ACCOUNT_HOLDER_NAME  VARCHAR(20)  NN
ADDRESS              LONGTEXT     NN
IFSC_CODE            VARCHAR(20)  NN UNIQUE CHECK(LENGTH(IFSC_CODE)=11)
LOC                  VARCHAR(30)  NN
BID                  INT UNSIGNED NN FOREIGN KEY
```

```
CREATE TABLE BANK
(
ACCOUNT_NUMBER INT NOT NULL PRIMARY KEY,
ACCOUNT_HOLDER_NAME VARCHAR(20) NOT NULL,
ADDRESS LONGTEXT NOT NULL,
IFSC_CODE VARCHAR(20) NOT NULL UNIQUE CHECK(LENGTH(IFSC_CODE)=11),
LOC VARCHAR(30) NOT NULL,
BID INT UNSIGNED NOT NULL,
CONSTRAINT FOREIGN KEY(BID)REFERENCES BRANCH(BID)
);
```

```
ALTER:
------
IT IS USED TO MODIFY THE TABLE STRUCTURE.
```

```
1.TO ADD NEW COLUMN_NAME
-------------------------
```

```
SYNTAX:
```

```
ALTER TABLE TABLE_NAME
ADD NEW_COLUMN_NAME NEW_DATATYPE NULL/NOT NULL;
```

```
BALANCE DECIMAL(5,2) NN-----STUDENT
```

```
ALTER TABLE STUDENT
ADD BALANCE DECIMAL(5,2) NOT NULL;
```

```
2.TO ADD NEW COLUMN_NAME AFTER PARTICULAR COLUMN
-----------------------------------------------
SYNTAX:
```

```
ALTER TABLE TABLE_NAME
ADD NEW_COLUMN_NAME NEW_DATATYPE NULL/NOT NULL AFTER EXISTING_COLUMN_NAME;
```

```
EMAIL VARCHAR(20) NULL-----STUDENT------PHONE
```

```
ALTER TABLE STUDENT
ADD EMAIL VARCHAR(20) NULL AFTER PHONE;
```

```
3.TO DROP COLUMN_NAME:
----------------------
SYNTAX:
```

```
ALTER TABLE TABLE_NAME
DROP EXISTING_COLUMN_NAME;
```

```
ALTER TABLE FACULTY
DROP FNAME;
```

```
4.TO CHANGE DATATYPE:
---------------------
SYNTAX:
```

```
ALTER TABLE TABLE_NAME
MODIFY EXISTING_COLUMN_NAME NEW_DATATYPE NULL/NOT NULL;
```

```
ALTER TABLE CUSTOMER
MODIFY CNAME CHAR(30) NOT NULL;
```

```
5.TO CHANGE NULL/NOT NULL
-------------------------
SYNTAX:
```

```
ALTER TABLE TABLE_NAME
MODIFY EXISTING_COLUMN_NAME EXISTING_DATATYPE NULL/NOT NULL;
```

```
ALTER TABLE BANK
MODIFY ADDRESS LONGTEXT NULL;
```

```
6.TO CHANGE TABLE_NAME:
```

```
------------------------
SYNTAX:

ALTER TABLE TABLE_NAME
RENAME NEW_TABLE_NAME;

BANK:BANK_DETAILS

ALTER TABLE BANK
RENAME BANK_DETAILS;

7.TO CHANGE COLUMN_NAME:
------------------------
SYNTAX:

ALTER TABLE TABLE_NAME
CHANGE EXISTING_COLUMN_NAME NEW_COLUMN_NAME EXISTING_DATATYPE NULL/NOT NULL;

LOC:LOCATION:BANK_DETAILS

ALTER TABLE BANK_dETAILS
CHANGE LOC LOCATION VARCHAR(30) NOT NULL;

ALTER TABLE STUDENT
CHANGE PHONE CONTACT_NUMBER BIGINT NOT NULL;

ERROR:WE CANT DROP/RENAME A COLUMN IF COLUMN ASSIGNED WITH CHECK CONSTRAINT.

TO ADD CONTRAINTS:
------------------
1.TO ADD PRIMARY KEY:
---------------------
SYNTAX:

ALTER TABLE TABLE_NAME
ADD PRIMARY KEY(COLUMN_NAME);

2.TO ADD UNIQUE:
-----------------
SYNTAX:

ALTER TABLE TABLE_NAME
ADD UNIQUE(COLUMN_NAME);

3.TO ADD CHECK:
---------------
SYNTAX:

ALTER TABLE TABLE_NAME
ADD CHECK(CONDITION);

(BALANCE>0)

ALTER TABLE STUDENT
ADD UNIQUE(EMAIL);

ALTER TABLE STUDENT
ADD CHECK(BALANCE>0);

4.TO ADD FOREIGN KEY:
---------------------
SYNTAX:

ALTER TABLE TABLE_NAME
ADD FOREIGN KEY(COLUMN_NAME)REFERENCES PARENT_TABLE_NAME(COLUMN_NAME);

ID INT NOT NULL----BANK_DETAILS
ID:FOREIGN KEY-----BANK_dETAILS

ALTER TABLE BANK_dETAILS
ADD ID INT NOT NULL;

ALTER TABLE BANK_dETAILS
ADD FOREIGN KEY(ID) REFERENCES STUDENT(ID);

5.TO AD DEFAULT CONSTRAINT:
-------------------------
SYNTAX:

ALTER TABLE TABLE_NAME
MODIFY EXISTING_COLUMN_NAME EXISTING_DATATYPE NULL/NOT NULL DEFAULT 'DEFAULT VALUE';

ALTER TABLE PRODUCT
MODIFY QUANTITY INT NULL DEFAULT '100';

6.TO ADD AUTO_INCREMENT:
------------------------
SYNTAX:
```

```
ALTER TABLE TABLE_NAME
MODIFY EXISTING_COLUMN_NAME EXISTING_DATATYPE NULL/NOT NULL AUTO_INCREMENT;

ALTER TABLE PRODUCT
MODIFY PID INT NOT NULL AUTO_INCREMENT;

TO DROP CONSTRAINTS:
--------------------
1.TO DROP PRIMARY KEY:
----------------------
ALTER TABLE TABLE_NAME
DROP PRIMARY KEY;

ALTER TABLE FACULTY
DROP PRIMARY KEY;

ALTER TABLE STUDENT
DROP PRIMARY KEY;

ERROR:WE CANT DROP PRIMARY KEY IF PRIMARY KEY COLUMN IS ACTING AS FOREIGN KEY IN ANOTHER TABLE.

ALTER TABLE PRODUCT
DROP PRIMARY KEY;

ERROR:WE CANT DROP PRIMARY KEY IF PRIMARY KEY COLUMN IS ASSIGNED WITH AUTO_INCREMENT.

2.TO DROP DEFAULT CONSTRAINT:
-----------------------------
SYNTAX:

ALTER TABLE TABLE_NAME
ALTER COLUMN COLUMN_NAME
DROP DEFAULT;

ALTER TABLE CUSTOMER
ALTER COLUMN BALANCE
DROP DEFAULT;

3.TO DROP AUTO_INCREMENT:
-------------------------
SYNTAX:

ALTER TABLE TABLE_NAME
MODIFY EXISTING_COLUMN_NAME EXISTING_DATATYPE NULL/NOT NULL;

ALTER TABLE CUSTOMER
MODIFY CID INT NOT NULL;

4.TO DROP UNIQUE,CHECK,FOREIGN KEY
----------------------------------
SYNTAX:

ALTER TABLE TABLE_NAME
DROP CONSTRAINT CONSTRAINT_NAME;

ALTER TABLE CUSTOMER
DROP CONSTRAINT EMAIL;

2.DML(DATA MANIPULATION LANGUAGE)
---------------------------------
1.INSERT
2.UPDATE
3.DELETE

1.INSERT:
---------
IT IS USED TO ADD/INSERT THE RECORDS INTO EXISTING TABLE.

SYNTAX:1(WHEN WE KNOW COLUMN_NAME AND COLUMN ORDER IN THE TABLE)
----------------------------------------------------------------

INSERT INTO TABLE_NAME VALUES(V1,V2,.......VN);

EX:

INSERT INTO PRODUCT VALUES(1,'BISCUIT',20,5);

TO RETRIEVE THE DATA FROM PRODUCT TABLE:
----------------------------------------
SELECT *
FROM PRODUCT;

INSERT INTO PRODUCT VALUES(2,'LIPSTICK',200,2);

INSERT INTO PRODUCT VALUES(3,'SHOES',500,3),(4,'DRESS',999,10);

INSERT INTO PRODUCT VALUES(5,'CRICKET BAT',999,2);
```

```
INSERT INTO PRODUCT VALUES(5,'CRICKET BAT CRICKET BALL',999,2);
ERROR:DATA TOO LONG FOR PNAME

INSERT INTO PRODUCT VALUES('PANT',6,300,2);ERROR

SYNTAX:2(WHEN WE KNOW ONLY COLUMN_NAME)
---------------------------------------

INSERT INTO TABLE_NAME(COL1,COL2,.....COLN)VALUES(V1,V2,.....VN);

INSERT INTO PRODUCT(PNAME,PRICE,PID,QUANTITY)VALUES('BELT',200,6,1);

INSERT INTO PRODUCT(PNAME,PRICE,PID,QUANTITY)VALUES
('DAIRY MILK',150,7,5),('KITKAT',40,8,10);

INSERT INTO PRODUCT(PNAME,PRICE)VALUES('LAPTOP',999);

CREATE TABLE EMPS
(
GENDER ENUM('M','F'));

INSERT INTO EMPS(GENDER)VALUES('M');

INSERT INTO EMPS(GENDER)VALUES('F');

INSERT INTO EMPS(GENDER)VALUES('O');ERROR

SYNTAX:3
---------

INSERT INTO TABLE_NAME(SELECT STATEMENT);

2.UPDATE:
---------
IT IS USED TO MODIFY EXISTING RECORDS IN A TABLE.

SYNTAX:
-------

UPDATE TABLE_NAME
SET COLUMN_NAME_1=V1,COLUMN_NAME_2=V2,........COLUMN_NAME_N=VN
[WHERE CONDITION];

1-----BIRIYANI
1-----CURD RICE

UPDATE PRODUCT
SET PNAME='BIRIYANI';

UPDATE PRODUCT
SET PNAME='CURD RICE'
WHERE PID=1;

5-----KABAB------100
5-----PANNEER----150

UPDATE PRODUCT
SET PNAME='PANNER',PRICE=150
WHERE PID=5;

 UPDATE PRODUCT
SET PNAME='APPLE',PRICE=50,QUANTITY=5
WHERE PID=6;

3.DELETE:
---------
IT IS USED TO DELETE EXISTING RECORDS FROM THE TABLE.

SYNTAX:

DELETE
FROM TABLE_NAME
[WHERE CONDITION];

DELETE
FROM PRODUCT
WHERE PID=9;

DDL:
----

3.TRUNCATE:
-----------
IT IS USED TO DELETE ALL THE RECORDS FROM THE TABLE WITHOUT AFFECTING TABLE STRUCTURE.

SYNTAX:
-------
TRUNCATE TABLE TABLE_NAME;
```

```
TRUNCATE TABLE PRODUCT;
```

4.DROP

IT IS USED TO DROP DATABASE AND ITS OBJECTS SUCH AS TABLE,VIEW,PROCEDURE,TRIGGER,FUNCTION AND USER FROM MYSQL..

i.TO DROP TABLE FROM DATABASE:

SYNTAX:

```
DROP TABLE TABLE_NAME;
```

```
DROP TABLE PRODUCT;
```

ii.TO DROP DATABASE FROM MYSQL

SYNTAX:

```
DROP DATABASE DATABASE_NAME;
```

DEMO

```
DROP DATABASE DEMO;
```

```
CREATE DATABASE ZOMATO;
```

```
USE ZOMATO;
```

1.WQTD EMPLOYEE FIRST NAME AND LAST NAME FROM EMPS TABLE???

```
SELECT fname,lname
FROM EMPS;
```

2.WQTD EMPLOYEE FIRST NAME,JOB,SALARY AND LID FROM EMPS TABLE???

```
SELECT FNAME,JOB,SAL,LID
FROM EMPS;
```

3.WQTD EMPLOYEE COMMISSION,DATE OF BIRTH,DATE OF JOINING AND MANAGER ID
FROM EMPS TABLE???

```
SELECT COMM,DOB,DOJ,MGR
FROM EMPS;
```

4.WQTD DETAILS OF EMPS FROM EMPS TABLE??

```
SELECT *
FROM EMPS;
```

5.WQTD FNAME,LAST NAME AND ANNUAL SALARY FROM EMPS TABLE????

```
SELECT FNAME,LNAME,SAL*12
FROM EMPS;
```

6.WQTD DETAILS OF EMPS ALONG WITH ANNUAL COMMISSION???

```
SELECT *,COMM*12
FROM EMPS;
```

7.WQTD FNAME,LNAME,JOB AND SALARY WITH 25000 BONUS???

```
SELECT FNAME,LNAME,JOB,SAL+25000
FROM EMPS;
```

8.WQTD FNAME,JOB,COMM AND SALARY WITH 10% HIKE??

```
SELECT FNAME,JOB,COMM,SAL+(SAL*10/100)
FROM EMPS;
```

HIKE:

```
COLUMN_NAME+(COLUM_NAME * %HIKE/100)
```

9.WQTD FNAME,LNAME,CID,LID AND COMM WITH 15% HIKE??

```
SELECT FNAME,LNAME,CID,LID,COMM+(COMM*15/100)
FROM EMPS;
```

10.WQTD FNAME,JOB AND ANNUAL SALARY WITH 8% HIKE???

```
SELECT FNAME,JOB,(SAL*12)+((SAL*12)*8/100)
FROM EMPS;
```

11.WQTD FNAME,LNAME,JOB AND SALARY WITH 5% DEDUCTION???

```
SELECT FNAME,LNAME,JOB,SAL-(SAL*5/100)
FROM EMPS;
```

DEDUCTION:

```
COLUMN_NAME-(COLUMN_NAME * %DEDUCTION/100)

12.WQTD FNAME,JOB,COMM WITH 12% HIKE AND ANNUAL SALARY WITH 20% DEDUCTION??

SELECT FNAME,JOB,COMM+(COMM*12/100),(SAL*12)-((SAL*12)*20/100)
FROM EMPS;

ALIASING:
---------
IT IS USED TO PROVIDE ALTERNATIVE NAME FOR A COLUMN IN RESULTANT TABLE.

RULES:
------
>WITH OR WITHOUT USING AS KEYWORD WE CAN WRITE ALIAS NAME.

>WE CAN USE MULTIPLE WORDS AS ALIAS NAME BY USING QUOTES OR BY
 CONNECTING IT WITH UNDERSCORE.

SELECT SAL AS SALARY
FROM EMPS;

SELECT SAL SALARY
FROM EMPS;

SELECT SAL*12 AS ANNUAL_SALARY
FROM EMPS;

SELECT SAL*12  'ANNUAL SALARY'
FROM EMPS;

SELECT FNAME,JOB,COMM+(COMM*12/100) AS 'COMMISSION WITH 12% HIKE',
(SAL*12)-((SAL*12)*20/100) 'ANNUAL SALARY WITH 20% DEDUCTION'
FROM EMPS;

COMMISSION WITH 12% HIKE
ANNUAL SALARY WITH 20% DEDUCTION

13.WQTD FNAME AS FIRST NAME,LNAME AS LAST NAME AND ANNUAL SALARY FROM EMPS
   TABLE???

SELECT FNAME 'FIRST NAME',LNAME 'LAST NAME',SAL*12 'ANNUAL SALARY'
FROM EMPS;

14.WQTD UNIQUE JOB ROLES PRESENT IN EMPS TABLE??

SELECT JOB
FROM EMPS;

DISTINCT:
---------
IT IS USED TO AVOID DUPLICATE VALUES FROM RESULTANT TABLE/
IT IS USED TO RETRIEVE UNIQUE VALUES FROM RESULTANT TABLE.

RULES:
------
>EITHER * OR DISTINCT MUST BE THE FIRST ARGUMENT IN SELECT CLAUSE.
(IF WE ARE USING SEPERATELY)

SELECT DISTINCT SAL,*
FROM EMPS;            O/P ERROR

SELECT *,DISTINCT SAL
FROM EMPS;            O/P ERROR

>WE CAN MULITPLE COLUMNS ALONG WITH DISTINCT,IT WILL AVOID COMBINATION OF DUPLICATE VALUE.

14.WQTD UNIQUE JOB ROLES PRESENT IN EMPS TABLE??

SELECT DISTINCT JOB
FROM EMPS;

15.WQTD DIFFERENT EMPLOYEE STATUS FROM EMPS TABLE??

SELECT DISTINCT STATUS
FROM EMPS;

16.WQTD UNIQUE COMBINATION OF EMPLOYEE STATUS AND MGR FROM EMPS TABLE?

SELECT DISTINCT STATUS,MGR
FROM EMPS;

SELECTION
---------
SYNTAX:

SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME
WHERE CONDITION;
```

```
where
-----
IT IS USED TO FILTER THE RECORDS FROM TABLE

Characteristics of where clause
-------------------------------
1. it executes after from clause
2. it executes row by row
3. it evaluates true or false conditions
4. we can write multiple condition inside where clause
5. we cant use alias name inside where clause

ORDER OF EXECUTION
------------------
1.FROM
2.WHERE
3.SELECT

17.WQTD DETAILS OF EMPS IF EMPS FNAME IS DIVYA???

SELECT *
FROM EMPS
WHERE FNAME='DIVYA';

18.WQTD FNAME AS FIRST NAME,LNAME AS LAST NAME ,JOB AND MGR IF
   EMPLOYEE IS WORKING AS WAITER??

SELECT FNAME 'FIRST NAME',LNAME AS 'LAST NAME',JOB,MGR
FROM EMPS
WHERE JOB='WAITER';

19.WQTD DETAILS OF EMPS WHO ARE GETTING SALARY MORE THAN 45000???

SELECT *
FROM EMPS
WHERE SAL>45000;

20.WQTD ALL THE DELIVERY EMPS DETAILS FROM EMPS TABLE??

SELECT *
FROM EMPS
WHERE JOB='DELIVERY';

21.WQTD DETAILS OF EMPS WHOSE STATUS IS BUSY???

SELECT *
FROM EMPS
WHERE  STATUS='BUSY';

22.WQTD FNAME,LNAME,DOB AND DOJ IF EMPLOYEE BORN AFTER THE YEAR 1993?

SELECT FNAME,LNAME,DOB,DOJ
FROM EMPS
WHERE DOB>'1993-12-31';
OR
WHERE DOB>='1994-01-01';

23.WQTD DETAILS OF EMPS WHO HIRED AFTER THE MONTH MARCH OF 2020???

SELECT *
FROM EMPS
WHERE DOJ>'2020-03-31';
OR
WHERE DOJ>='2020-04-01';

24.WQTD DETAILS OF EMPS WHO JOINED BEFORE THE YEAR 2019?

SELECT *
FROM EMPS
WHERE DOJ<'2019-01-01';
OR
WHERE DOJ<='2018-12-31';

25.WQTD DETAILS OF EMPS ALONG WITH ANNUAL SALARY IF EMPS ARE GETTING
   ANNUAL SALARY MORE THAN 500000???

SELECT *,SAL*12 ANNUAL_SALARY
FROM EMPS
WHERE (SAL*12)>500000;

OPERATORS:
----------
1.ARITHMATIC OPERATORS:(+,-,*,/,%)
2.RELATIONAL OPERATORS:(>,<,>=,<=,=,!=)
3.LOGICAL OPERATORS   :(AND,OR,NOT)
```

```
4.SPECIAL OPERATORS   :(IN,NOT IN,IS,BETWEEN,NOT BETWEEN,LIKE,NOT LIKE)
5.SUBQUERY OPERATORS  :(ALL,ANY)

26.WQTD DETAILS OF EMPS IF EMPS ARE WORKING AS CHEF OR
   CASHIER??

SELECT *
FROM EMPS
WHERE JOB='CHEF' OR JOB='CASHIER';

27.WQTD FNAME,LNAME,JOB,DOB AND DOJ IF EMPS BORN DURING
   THE YEAR 1995???

SELECT FNAME,LNAME,JOB,DOB,DOJ
FROM EMPS
WHERE DOB>='1995-01-01' AND DOB<='1995-12-31';

AISHWARYA:1995-10-12   T AND  T:T
SHRIHARI :1996-12-31   T AND  F:F

28.WQTD DETAILS OF THE EMPS IF EMPS ARE WORKING AS SECURITY AND GETTING SALARY LESS THAN 80000??

SELECT *
FROM EMPS
WHERE JOB='SECURITY' AND SAL<80000;

29.WQTD DETAILS OF EMPS WHO ARE WORKING AS SECURITY OR MANAGER
   AND THEIR STATUS IS AVAILABLE???

SELECT *
FROM EMPS
WHERE (JOB='SECURITY' OR JOB='MANAGER') AND (STATUS='AVAILABLE');

30.WQTD DETAILS OF EMPS IF EMPS ARE WORKING AS SECURITY,CHEF,DELIVERY OR WAITER AND GETTING SALARY MORE
THAN 45000 AND HIRED IN THE YEAR 2018???

SELECT *
FROM EMPS
WHERE (JOB='SECURITY' OR JOB='CHEF' OR JOB='DELIVERY' OR JOB='WAITER') AND (SAL>45000) AND (DOJ>='2018-
01-01' AND DOJ<='2018-12-31');

IN:
----
IT IS A MULTI VALUE OPERATOR WHICH TAKES MULTIPLE VALUES AT THE RHS AND SINGLE VALUE AT THE LHS.

SYNTAX:
-------
            LHS                RHS
COLUMN_NAME/EXPRESSION IN(V1,V2,.....VN);

>IT WORKS ON OR CONDITION

30.WQTD DETAILS OF EMPS IF EMPS ARE WORKING AS SECURITY,CHEF,DELIVERY OR WAITER AND GETTING SALARY MORE
THAN 45000 AND HIRED IN THE YEAR 2018???

SELECT *
FROM EMPS
WHERE JOB IN('SECURITY','CHEF','DELIVERY','WAITER')
AND (SAL>45000) AND (DOJ>='2018-01-01' AND DOJ<='2018-12-31');

31.WQTD DETAILS OF THE ITEMS FROM MENU_ITEMS TABLE IF ITEM NAME IS MASALA DOSA OR FISH CURRY AND PRICE
IS LESS THAN 500??

SELECT *
FROM MENU_ITEMS
WHERE NAME IN('MASALA DOSA','FISH CURRY') AND PRICE<500;

32.WQTD RESTAURANT NAME ,ADDRESS,CITY IF RESTAURANT IS LOCATED
   EITHER IN BANGALORE OR MUMBAI CITY IN THE ADDRESS MG ROAD??

SELECT NAME,ADDRESS,CITY
FROM RESTAURANTS
WHERE CITY IN('BANGALORE','MUMBAI') AND ADDRESS='MG ROAD';

33.WQTD DETAILS OF EMPS WHO ARE NOT WORKING AS CHEF,CLEANER
    AND CASHIER(WITHOUT USING SPECIAL OPERATOR)

SELECT *
FROM EMPS
WHERE JOB!='CHEF' AND JOB!='CLEANER' AND JOB!='CASHIER';

NOT IN
------
IT IS A MULTI VALUE OPERATOR WHICH TAKES MULTIPLE VALUES AT THE RHS AND SINGLE VALUE AT THE LHS.

SYNTAX:
-------
```

```
          LHS                    RHS
COLUMN_NAME/EXPRESSION NOT IN(V1,V2,....V3);
```

>IT WORKS ON AND CONDITION.

33.WQTD DETAILS OF EMPS WHO ARE NOT WORKING AS CHEF,CLEANER
AND CASHIER(WITHOUT USING SPECIAL OPERATOR)

```
SELECT *
FROM EMPS
WHERE JOB NOT IN('CHEF','CLEANER','CASHIER');
OR
WHERE JOB!='CHEF' AND JOB!='CLEANER' AND JOB!='CASHIER';
```

34.WQTD DETAILS OF EMPS WHOSE SALARY IS MORE THAN OR EQUAL TO 35000 AND LESS THAN OR EQUAL TO 50000 AND NOT WORKING AS MANAGER,CHEF??

```
SELECT *
FROM EMPS
WHERE SAL BETWEEN 35000 AND 50000 AND JOB NOT IN('MANAGER','CHEF');
OR
WHERE SAL>=35000 AND SAL<=50000 AND JOB NOT IN('MANAGER','CHEF');
```

```
BETWEEN:
--------
IT IS USED TO INCLUDE SOME RANGE OF VALUES.

SYNTAX:
-------
                                      >=                  <=
COLUMN_NAME/EXPRESSION BETWEEN LOWER_RANGE_VALUE AND HIGHER_RANGE_VALUE;
```

35.WQTD DETAILS OF EMPS WHO ARE WORKING AS WAITER,CLEANER OR CASHIER AND
GETTING SALARY MORE THAN 20000 AND LESS THAN 40000???

```
SELECT *
FROM EMPS
WHERE JOB IN('WAITER','CLEANER','CASHIER') AND SAL BETWEEN 20000+0.01  AND 40000-0.01 ;

SAL decimal(10,2)   : 99999999.99
                    : 0.01
```

36.WQTD DETAILS OF EMPS WHO WERE BORN IN THE YEAR 1995??

```
SELECT *
FROM EMPS
WHERE DOB BETWEEN '1995-01-01' AND '1995-12-31';
```

37.WQTD FNAME,JOB,DOB AND DOJ IF EMPS ARE WORKING AS WAITER OR DELIVERY BUT NOT WORKING AS SECURITY AND CLEANER AND HIRED IN THE YEAR 2018 OR 2019??

```
SELECT FNAME,JOB,DOB,DOJ
FROM EMPS
WHERE JOB IN('WAITER','DELIVERY') AND JOB NOT IN('SECURITY','CLEANER') AND DOJ BETWEEN '2018-01-01' AND
'2019-12-31';
```

38.WQTD DETAILS OF EMPS WHO ARE NOT GETTING SALARY IN THE RANGE OF 35000 TO
45000???

```
SELECT *
FROM EMPS
WHERE SAL NOT BETWEEN 35000 AND 45000;
```

```
NOT BETWEEN:
-----------
IT IS USED TO EXCLUDE SOME RANGE OF VALUES.

SYNTAX:
-------
                                          <                 >
COLUMN_NAME/EXPRESSION NOT BETWEEN LOWER_RANGE_VALUE AND HIGHER_RANGE_VALUE;
```

39.WQTD DETAILS OF EMPS WHO WERE NOT JOINED IN THE YEAR 2020??

```
SELECT *
FROM EMPS
WHERE DOJ NOT BETWEEN '2020-01-01' AND '2020-12-31';
```

40.WQTD DETAILS OF EMPS WHO ARE GETTING SOME COMMISSION??

```
SELECT *
FROM EMPS
WHERE COMM IS NOT NULL;

IS:
----
```

IT IS USED TO CHECK WHETHER COLUMN IS NULL/NOT NULL.

SYNTAX:

```
            LHS                RHS
COLUMN_NAME/EXPRESSION IS NULL/NOT NULL;
```

```
SELECT *
FROM EMPS
WHERE COMM IS NULL;
```

41.WQTD DETAILS OF EMPS WHO ARE HAVING REPORTING MANAGER???

```
SELECT *
FROM EMPS
WHERE MGR IS NOT NULL;
```

42.WQTD DETAILS OF EMPS WHO ARE ACTING AS CUSTOMER FOR THEIR COMPANY?

```
SELECT *
FROM EMPS
WHERE CID IS NOT NULL;
```

```
SELECT *
FROM EMPS
WHERE CID IS NULL;
```

```
SELECT *
FROM EMPS
WHERE FNAME IS 'PRIYA';ERROR
```

43.WQTD DETAILS OF EMPS WHO JOINED IN THE YEAR 2016 AND IN THE DATE 10?

```
SELECT *
FROM EMPS
WHERE DOJ IN('2016-01-10','2016-02-10','2016-03-10','2016-04-10','2016-05-10','2016-06-10','2016-07-10',
'2016-08-10','2016-09-10','2016-10-10','2016-11-10','2016-12-10');
```

44.WQTD DETAILS OF EMPS WHO ARE WORKING AS WAITER OR DELIVERY AND GETTING SALARY IN THE RANGE OF 35000 TO 100000 AND HIRED IN THE YEAR 2018 AND MUST HAVE A REPORTING MANAGER???

```
SELECT *
FROM EMPS
WHERE JOB IN('WAITER','DELIVERY') AND SAL BETWEEN 35000 AND 100000 AND DOJ BETWEEN '2018-01-01' AND
'2018-12-31' AND MGR IS NOT NULL;
```

45.WQTD DETAILS OF EMPS IF FNAME IS STARTING WITH A???

```
select *
FROM EMPS
WHERE FNAME LIKE 'A%';
```

46.WQTD FNAME AND LNAME IF LNAME IS ENDING WITH Y?

```
SELECT FNAME,LNAME
FROM EMPS
WHERE LNAME LIKE '%Y';
```

47.WQTD FNAME,LNAME,SALARY AND JOB IF EMPS FNAME CONTAINS ATLEAST 2 A AND GETTING SALARY IN THE RANGE OF 40000 TO 100000??

```
SELECT FNAME,LNAME,SAL,JOB
FROM EMPS
WHERE FNAME LIKE '%A%A%' AND SAL BETWEEN 40000 AND 100000;
```

%A%A%A%

48.WQTD FNAME,LNAME AND JOB IF JOB CONTAINS A STRING ER IN IT??

```
SELECT FNAME,LNAME,JOB
FROM EMPS
WHERE JOB LIKE '%ER%';
```

49.WQTD DETAILS OF EMPS WHOSE FNAME STARTING WITH S OR A???

```
SELECT *
FROM EMPS
WHERE FNAME LIKE 'S%' OR FNAME LIKE 'A%';
```

50.WQTD DETAILS OF EMPS IF EMPS HIRED IN THE YEAR 2019???

```
SELECT *
FROM EMPS
WHERE DOJ LIKE '2019%';
```

2019-%-%

YYYY-MM-DD

51.WQTD DETAILS OF EMPS WHO JOINED IN THE DATE 12 OR 01??

```
SELECT *
FROM EMPS
WHERE DOJ LIKE '%12' OR DOJ LIKE '%01';
```

52.WQTD FNAME,LNAME,DOB IF EMPS BORN IN THE MONTH OF JAN,FEB OR
MARCH???

```
SELECT FNAME,LNAME,DOB
FROM EMPS
WHERE DOB LIKE '%-01-%' OR DOB LIKE '____-02-__' OR DOB LIKE '%-03-__';
```

%01% %02% %03%

```
A:1999-10-01  F
B:2000-01-11  T    T
C:2001-12-15  F
```

YYYY-MM-DD

53.WQTD DETAILS OF THE EMPS ALONG WITH ANNUAL SALARY IF ANNUAL SALRY 2ND DIGIT IS 2??

```
select *,SAL*12 AS 'ANNUAL SALARY'
FROM EMPS
WHERE SAL*12 LIKE '_2%';
```

54.WQTD DETAILS OF EMPS WHOSE FNAME NOT STARTING WITH D???

```
NOT LIKE:
---------
IT IS USED FOR PATTERN MATCHING.
```

>IT WILL EXCLUDE THE DETAILS BASED ON THE PATTERN.

```
SYNTAX:
-------
```

```
COLUMN_NAME/EXPRESSION NOT LIKE 'PATTERN_TO_MATCH';
```

```
SELECT *
FROM EMPS
WHERE FNAME NOT LIKE 'D%';
```

55.WQTD DETAILS OF EMPS IF EMPS WERE NOT JOINED IN THE YEAR 2020??

```
SELECT *
FROM EMPS
WHERE DOJ NOT LIKE '2020%';
```

56.WQTD DETAILS OF EMPS WHOSE FNAME NOT STARTING WITH S AND D???

```
SELECT *
FROM EMPS
WHERE FNAME NOT LIKE 'S%' AND FNAME NOT LIKE 'D%';
```

57.WQTD FNAME,LNAME IF LNAME CONTAINS EXACTLY 3 CHARACTERS??

```
SELECT FNAME,LNAME
FROM EMPS
WHERE LNAME LIKE '___';
```

58.WQTD DETAILS OF EMPS IF EMPLOYEE FNAME LAST 3RD CHARACTER IS E AND LNAME LAST CHARACTER IS R AND
WORKING AS WAITER OR MANAGER BUT NOT AS FEMALE??

```
SELECT *
FROM EMPS
WHERE FNAME LIKE '%E__' AND LNAME LIKE '%R' AND JOB IN('WAITER','MANAGER') AND GENDER!='F';
```

59.WQTD DETAILS OF EMPS IF EMPS FNAME LAST 5TH CHARACTER IS H AND EMPS JOB ROLE 3RD CHARACTER IS I AND
EMPLOYEE IS NOT ACTING AS CUSTOMER FOR HIS COMPANY???

```
SELECT *
FROM EMPS
WHERE FNAME LIKE '%H____' AND JOB LIKE '__I%' AND CID IS NULL;
```

60.WQTD DETAILS OF EMPS WHOSE FNAME IS STARTING VOWELS???

```
SELECT *
FROM EMPS
WHERE FNAME LIKE 'A%' OR
      FNAME LIKE 'E%' OR
      FNAME LIKE 'I%' OR
      FNAME LIKE 'O%' OR
      FNAME LIKE 'U%';
```

61.WQTD FNAME,LNAME OF EMPS IF FNAME IS STARTING WITH VOWELS AND LNAME
IS NOT ENDING WITH VOWELS??

```
SELECT FNAME,LNAME
FROM EMPS
WHERE (FNAME LIKE 'A%' OR
      FNAME LIKE 'E%' OR
      FNAME LIKE 'I%' OR
      FNAME LIKE 'O%' OR
      FNAME LIKE 'U%') AND
      (LNAME NOT LIKE '%A' AND
      LNAME NOT LIKE '%E' AND
      LNAME NOT LIKE '%I' AND
      LNAME NOT LIKE '%O' AND
      LNAME NOT LIKE '%U');
```

62.WQTD NAME OF EMPS FROM PENTAGON TABLE IF NAME CONTAINS ATLEAST 1 % IN IT??

```
CREATE TABLE PENTAGON
(
ID INT,
NAME VARCHAR(20)
);
```

```
INSERT INTO TABLE_NAME VALUES(V1,V2,....VN);
```

```
INSERT INTO PENTAGON VALUES(1,'RAKSH%ITH'),(2,'JUL_IET'),(3,'RAD%H%A');
```

```
SELECT NAME
FROM PENTAGON
WHERE NAME LIKE '%\%%';
```

\ IT IS USED TO REMOVE SPECIAL BEHAVIOUR FROM SPECIAL CHARACTER.

63.WQTD NAME OF EMPS IF NAME CONTAINS ATLEAST 2 % IN IT???

```
SELECT NAME
FROM PENTAGON
WHERE NAME LIKE '%\%%\%%';
```

64.WQTD NAME OF EMPS IF NAME 4TH CHARACTER IS _

```
SELECT NAME
FROM PENTAGON
WHERE NAME LIKE '___\_%';
```

```
FUNCTION:
---------
1.AGGREGATE FUNCTION/GROUP FUNCTION
------------------------------------
TYPES OF AGGREGATE FUNCTIONS:
-----------------------------
1.MAX():
--------
IT IS USED TO OBTAIN MAXIMUM VALUE FROM THE GIVEN COLUMN.
```

SYNTAX: MAX(COLUMN_NAME/EXPRESSION)

EX:

```
SELECT MAX(SAL)
FROM EMPS;
```

O/P:100000.00

```
2.MIN():
--------
IT IS USED TO OBTAIN MINIMUM VALUE FROM THE GIVEN COLUMN.
```

SYNTAX: MIN(COLUMN_NAME/EXPRESSION)

EX:

```
SELECT MIN(SAL)
FROM EMPS;
```

O/P:20000.00

```
3.SUM():
--------
IT IS USED TO OBTAIN TOTAL VALUE  FROM THE GIVEN COLUMN.
```

SYNTAX: SUM(COLUMN_NAME/EXPRESSION)

EX:

```
SELECT SUM(SAL)
FROM EMPS;
```

O/P:510000.00

4.AVG():

IT IS USED TO OBTAIN AVERAGE VALUE FROM THE GIVEN COLUMN.

SYNTAX: AVG(COLUMN_NAME/EXPRESSION)

EX:

```
SELECT AVG(SAL)
FROM EMPS;
```

O/P:51000.00

5.COUNT():

IT IS USED TO OBTAIN NUMBER OF VALUES PRESENT IN GIVEN COLUMN.

SYNTAX: COUNT(*/COLUMN_NAME/EXPRESSION)

EX:

```
SELECT COUNT(SAL)
FROM EMPS;
```

O/P:10

```
SELECT COUNT(*)
FROM EMPS;
```

O/P:10

NOTE:ONLY FOR COUNT() WE CAN USE * AS A ARGUMENT.

CHARACTERISTICS OF AGGREGATE FUNCTIONS:

>IT TAKES N NUMBER OF INPUTS AND GENERATE SINGLE O/P.
>IT EXECUTES GROUP BY GROUP.
>WE CANT USE NORMAL COLUMNS ALONG WITH AGGREGATE FUNCTIONS IN
 SELECT CLAUSE
>WE CAN USE ONLY ONE COLUMN AS A ARGUMENT FOR AGGREGATE
 FUNCTIONS.
>WE CANT USE AGGREGATE FUNCTIONS INSIDE WHERE CLAUSE.
>WE CANT NEST AGGREAGET FUNCTIONS.
>IT IGNORES NULL VALUES.
>WE CAN USE GROUP BY EXPRESSION ALONG WITH AGGREGATE FUNCTIONS
 IN SELECT CLAUSE.
65.WQTD TOTAL SALARY GIVEN TO ALL THE WAITERS??

```
SELECT SUM(SAL)
FROM EMPS
WHERE JOB='WAITER';
```

66.WQTD AVERAGE SALARY,TOTAL SALARY,MINIMUM SALARY AND MAXIMUM SALARY IF EMPS JOB 3RD CHARACTER IS I OR C???

```
SELECT AVG(SAL),SUM(SAL),MIN(SAL),MAX(sAL)
FROM EMPS
WHERE JOB LIKE '__I%' OR JOB LIKE '__C%';
```

67.WQTD NUMBER OF EMPS WORKING AS DELIVERY,WAITER OR MANAGER AND GETTING SALARY MORE THAN 30000 AND LESS THAN 100000?

```
SELECT COUNT(*)
FROM EMPS
WHERE JOB IN('DELIVERY','WAITER','MANAGER') AND SAL BETWEEN 30000+0.01 AND 100000-0.01;
```

0.01

68.WQTD NUMBER OF UNIQUE JOB ROLES FROM EMPS TABLE??

```
SELECT COUNT(DISTINCT JOB)
FROM EMPS;
```

69.WQTD NUMBER OF EMPS WORKING IN EACH JOB ROLE???

```
select COUNT(*),JOB
FROM EMPS
GROUP BY JOB;
```

70.WQTD STATUS WISE NO OF EMPS WORKING IN EMPS TABLE???

```
SELECT COUNT(*),STATUS
FROM EMPS
GROUP BY STATUS;
```

71.WQTD TOTAL SALARY SPENT BY THE COMPANY IN EACH JOB ROLE IF

JOB ROLES ARE DELIVERY,MANAGER OR WAITER???

```
SELECT SUM(SAL),JOB
FROM EMPS
WHERE JOB IN('DELIVERY','MANAGER','WAITER')
GROUP BY JOB;
```

72.WQTD NUMBER OF EMPS WORKING IN EACH LOCATION IF EMPS ARE GETTING SALARY MORE THAN 32000 AND LESS THAN 50000???

```
SELECT COUNT(*),LID
FROM EMPS
WHERE SAL>32000 AND SAL<50000
GROUP BY LID;
```

73.WQTD MAXIMUM SALARY OBTAINED PER LOCATION IF EMPS ARE GETTING SALARY MORE THAN 25000 AND MAXIMUM SALARY OBTAINED IN THE LOCATION IS MORE THAN 50000?

```
SELECT MAX(SAL),LID
FROM EMPS
WHERE SAL>25000
GROUP BY LID
HAVING MAX(SAL)>50000;
```

```
ORDER OF EXECUTION:
-------------------
1.FROM
2.[WHERE]
3.GROUP BY
4.HAVING
5.SELECT
```

74.WQTD TOTAL SALARY AND NUMBER OF EMPS WORKING IN EACH LOCATION IF LOCATION CONTAINS ATLEAST 2 EMPS AND TOTAL SALARY IS MORE THAN 40000??

```
SELECT SUM(SAL),COUNT(*),LID
FROM EMPS
GROUP BY LID
HAVING COUNT(*)>=2 AND SUM(sAL)>40000;
```

75.WQTD NUMBER OF EMPS ,MAXIMUM SALARY AND MINIMUM SALARY OBTAINED IN EACH LOCATION IF LID IS 2,3,4 OR 5 AND NUMBER OF EMPS WORKING IN LOCATION IS ATLEAST 1 OR ATMOST 2???

```
SELECT COUNT(*),MAX(sAL),MIN(SAL),LID
FROM EMPS
WHERE LID IN(2,3,4,5)
GROUP BY LID
HAVING COUNT(*)>=1 OR COUNT(*)<=2;
```

76.WQTD STATUS WISE TOTAL SALARY AND NUMBER OF EMPS WORKING IN EMPS TABLE IF STATUS 3RD CHARACTER IS S OR A AND TOTAL SALARY OBTAINED IS MORE THAN 70000???

```
SELECT SUM(SAL),COUNT(*),STATUS
FROM EMPS
WHERE STATUS LIKE '__S%' OR STATUS LIKE '__A%'
GROUP BY STATUS
HAVING SUM(sAL)>70000;
```

77.WQTD NUMBER OF EMPS WHO ARE GETTING SAME SALARIES/REPEATED SALARY?

```
SELECT COUNT(*),SAL
FROM EMPS
GROUP BY SAL
HAVING COUNT(*)>1;
```

78.WQTD NUMBER OF EMPS HAVING SAME GENDER AND WORKING IN SAME JOB ROLE??

```
SELECT COUNT(*),GENDER,JOB
FROM EMPS
GROUP BY GENDER,JOB
HAVING COUNT(*)>1;
```

79.WQTD DETAILS OF EMPS BASED ON THEIR SALARY MAXIMUM TO MINIMUM ORDER??

```
ORDER BY:
---------
IT IS USED TO ARRENGE THE RECORDS EITHER IN ASCENDING OR DESCENDING ORDER.

SYNTAX:
-------
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME
ORDER BY COLUMN_NAME ASC/DESC;
```

CHARACTERISTICS OF ORDER BY:

>IT IS A LAST EXECUTABLE CLAUSE IN A QUERY.
>IT EXECUTES AFTER SELECT CLAUSE.
>BY DEFAULT IT WILL CONSIDER ASCENDING ORDER FOR THE COLUMN.
>NORMALLY IN ALL THE TABLES THE RECORDS ARE ARRENGED IN ASCENDING
ORDER BASED ON PRIMARY KEY COLUMN.
>WE CAN USE ALIAS NAME INSIDE ORDER BY CLAUSE.
>WE CAN USE MULTIPLE COLUMNS INSIDE ORDER BY CLAUSE.
(IT WILL GIVE THE PRIORITY FOR FIRST COLUMN,IF VALUES ARE SAME IN
THE FIRST COLUMN THEN IT WILL GIVE THE PRIORITY FOR 2ND COLUMN)

ORDER OF EXECUTION:

1.FROM
2.WHERE
3.GROUP BY
4.HAVING
5.SELECT
6.ORDER BY

79.WQTD DETAILS OF EMPS BASED ON THEIR SALARY MAXIMUM TO MINIMUM
ORDER??

```
SELECT *
FROM EMPS
ORDER BY SAL DESC;
```

80.WQTD FNAME,LNAME AND JOB IF EMPS ARE WORKING AS SECURITY,MANAGER OR CLEANER AND ARRENGE THE RECORDS
IN ALPHABATICAL ORDER BASED ON THEIR FNAME???

```
SELECT FNAME,LNAME,JOB
FROM EMPS
WHERE JOB IN('SECURITY','MANAGER','CLEANER')
ORDER BY FNAME ASC;
```

81.WQTD NUMBER OF EMPS WHO ARE HAVING SAME GENDER AND WORKING IN SAME JOB ROLE IF EMPS JOB ROLE 4TH
CHARACTER IS I OR T AND ARRENGE THE JOB IN ANTI ALPHABATICAL ORDER???

```
SELECT COUNT(*),GENDER,JOB
FROM EMPS
WHERE JOB LIKE '___I%' OR JOB LIKE '___T%'
GROUP BY GENDER,JOB
HAVING COUNT(*)>1
ORDER BY JOB DESC;
```

82.WQTD DETAILS OF FIRST 3 RECORDS FROM EMPS TABLE??

LIMIT:

IT IS USED TO PRINT SOME SPECIFIC NUMBER OF RECORDS.

SYNTAX:

```
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME
LIMIT VALUE;
```

```
SELECT *
FROM EMPS
LIMIT 3;
```

83.WQTD DETAILS OF FIRST RECORD FROM EMPS TABLE???

```
SELECT *
FROM EMPS
LIMIT 1;
```

84.WQTD DETAILS OF 2ND RECORD FROM EMPS TABLE??

```
SELECT *
FROM EMPS
LIMIT 1 OFFSET 1;
```

OFFSET:

IT IS USED TO SKIP/IGNORE SOME SPECIFIC NUMBER OF RECORDS.

SYNTAX:

```
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME
LIMIT VALUE OFFSET VALUE;
```

```
SELECT *
FROM EMPS
LIMIT 1 OFFSET 1;
```

85.WQTD DETAILS OF 4TH AND 5TH RECORD FROM EMPS TABLE???

```
SELECT *
FROM EMPS
LIMIT 2 OFFSET 3;
```

86.WQTD DETAILS OF 3RD AND 5TH RECORD FROM EMPS TABLE???

```
SELECT *
FROM EMPS
LIMIT 2 OFFSET 3;
```

UNION

87.WQTD TOP 3 HIGHEST PAID EMPS DETAILS???

```
SELECT *
FROM EMPS
ORDER BY SAL DESC
LIMIT 3;
```

88.WQTD TOP 5 LOWEST PAID EMPS DETAILS??

```
SELECT *
FROM EMPS
ORDER BY SAL ASC
LIMIT 5;
```

89.WQTD LAST 4 RECORDS FROM EMPS TABLE???

```
SELECT *
FROM EMPS
ORDER BY EID DESC
LIMIT 4;
```

90.WQTD 3RD MAXIMUM SALARY FROM EMPS TABLE???

```
SELECT DISTINCT SAL
FROM EMPS
ORDER BY SAL DESC
LIMIT 1 OFFSET 2;
```

OR

```
SELECT SAL
FROM EMPS
GROUP BY SAL
ORDER BY SAL DESC
LIMIT 1 OFFSET 2;
```

91.WQTD 4TH MINIMUM SALARY FROM EMPS TABLE???

```
SELECT DISTINCT SAL
FROM EMPS
ORDER BY SAL ASC
LIMIT 1 OFFSET 3;
```

CHARACTER FUNCTIONS:

CHARACTERISTICS OF CHARACTER FUNCTIONS:

>IT TAKES N NUMBER OF INPUTS AND GENERATE N NUMBER OF OUTPUTS.
>IT EXECUTES ROW BY ROW
>WE CAN NEST CHARACTER FUNCTIONS.
>WE CAN USE CHARACTER FUNCTIONS INSIDE WHERE CLAUSE.

TYPES OF CHARACTER FUNCTIONS:

1.LOWER():

IT IS USED TO CONVERT STRING VALUE INTO LOWERCASE.

EX:

```
SELECT LOWER('SHEELA');
```

O/P:sheela

2.UPPER():

IT IS USED TO CONVERT STRING VALUE INTO UPPERCASE.

EX:

```
SELECT UPPER('manja');
```

o/p:MANJA

3.LENGTH():

IT IS USED TO OBTAIN TOTAL NUMBER OF CHARACTERS PRESENT IN GIVEN STRING VALUE.

EX:

```
SELECT LENGTH('PENTAGON');8

SELECT LENGTH('PENTAGON SPACE');14
```

92.WQTD 2ND LONGEST CITY AS WELL AS THEIR RESPECTIVE LENGTH FROM LOCATIONS TABLE,IF THERE IS MORE THAN 1 2ND LONGEST CITY CHOOSE THE ONE THAT COMES FIRST WHEN WE PLACED IN ALPHABATICAL ORDER??

```
SELECT CITY,LENGTH(CITY)
FROM LOCATIONS
ORDER BY LENGTH(CITY) DESC,CITY ASC
LIMIT 1 OFFSET 1;

SELECT CITY,LENGTH(CITY)
FROM LOCATIONS
ORDER BY LENGTH(CITY) ASC,CITY ASC
LIMIT 1 OFFSET 1;
```

4.CONCAT():

IT IS USED TO ADD/COMBINE STRING VALUES.

SYNTAX:

```
CONCAT('STR1','STR2',.......'STRN');
```

SAMPLE O/P:

```
MR/MISS SHAKILA YOUR SALARY IS 420 RS.
        FNAME                  SAL

SELECT CONCAT('MR/MISS ',FNAME,' YOUR SALARY IS ',SAL,' RS.')
FROM EMPS;
```

93.WQTD FNAME AND LNAME TOGETHER AS FULL_NAME FROM EMPS TABLE??

I/P:

FNAME:KICCHA LNAME:SUDEEP

O/P:

```
FULL_NAME:
----------
KICCHA SUDEEP

SELECT CONCAT(FNAME,' ', LNAME)AS FULL_NAME
FROM EMPS;
```

94.WQTD FNAME,LNAME AND JOB TOGETHER IN BELOW FORMAT:

```
O/P:
----
SHARUKH KHAN (HERO)

FNAME:SHARUKH
LNAME:KHAN
JOB:HERO

SELECT CONCAT(FNAME,' ',LNAME,' ','(',JOB,')')
FROM EMPS;
```

5.REVERSE():

IT IS USED TO DISPLAY GIVEN STRING IN REVERSE FORMAT.

EX:

```
SELECT REVERSE('DBOSS');SSOBD
```

6.SUBSTR():

IT IS USED EXTRACT SOME PART OF THE STRING FROM ORIGINAL STRING.

SYNTAX: SUBSTR('ORIGINAL STRING',POSITION,[LENGTH]);

CASE:1

```
 K I P I K E E R T H  I
 1 2 3 4 5 6 7 8 9 10 11
```

```
SELECT SUBSTR('KIPIKEERTHI',5,3);KEE
```

```
SELECT SUBSTR('KIPIKEERTHI',2,6);IPIKEE
```

```
SELECT SUBSTR('KIPIKEERTHI',10);HI
```

```
SELECT SUBSTR('KIPIKEERTHI',8);RTHI
```

```
CASE:2
```

```
 K  I  P I K E E R T H I
-11-10-9-8-7-6-5-4-3-2-1
```

```
SELECT SUBSTR('KIPIKEERTHI',-4,3);RTH
SELECT SUBSTR('KIPIKEERTHI',-6);EERTHI
```

```
96.WQTD FNAME AND LNAME IF FNAME IS STARTING WITH K??
```

```
SELECT FNAME,LNAME
FROM EMPS
WHERE SUBSTR(FNAME,1,1)='K';
```

```
97.WQTD FNAME AND LNAME IF LNAME IS ENDING WITH I??
```

```
SELECT FNAME,LNAME
FROM EMPS
WHERE SUBSTR(LNAME,-1,1)='I';
```

```
98.WQTD DETAILS OF EMPS WHOSE FNAME STARTING WITH A OR D???
```

```
SELECT *
FROM EMPS
WHERE SUBSTR(FNAME,1,1) IN('A','D');
OR
WHERE SUBSTR(FNAME,1,1)='A' OR SUBSTR(FNAME,1,1)='D';
OR
WHERE FNAME LIKE 'A%' OR FNAME LIKE 'D%';
```

```
99.WQTD FNAME AND JOB IF JOB STARTS WITH WAIT OR SEC???
```

```
SELECT FNAME,JOB
FROM EMPS
WHERE SUBSTR(JOB,1,4)='WAIT' OR SUBSTR(JOB,1,3)='SEC';
```

```
100.WQTD DETAILS OF EMPS ALONG WITH ANNUAL SALARY IF ANNUAL SALARY 2ND DIGIT IS 2 AND JOB IS NOT ENDING
WITH ER??
```

```
SELECT *,SAL*12 AS 'ANNUAL SALARY'
FROM EMPS
WHERE SUBSTR(SAL*12,2,1)=2 AND SUBSTR(JOB,-2,2)!='ER';
```

```
101.WQTD FNAME AND LNAME IF FNAME IS STARTING WITH VOWEL AND LNAME IS NOT ENDING WITH VOWEL??
```

```
SELECT FNAME,LNAME
FROM EMPS
WHERE SUBSTR(FNAME,1,1)IN('A','E','I','O','U') AND SUBSTR(LNAME,-1,1) NOT IN('A','E','I','O','U');
```

```
102.WQTD DETAILS OF EMPS WHO BORN IN THE YEAR 1995???
```

```
SELECT *
FROM EMPS
WHERE SUBSTR(DOB,1,4)=1995;
```

```
YYYY-MM-DD
12345678910
```

```
103.WQTD DETAILS OF EMPS HIRED IN THE MONTH APRIL,MAY,NOVEMBER OR DECEMBER??
```

```
SELECT *
FROM EMPS
WHERE SUBSTR(DOJ,6,2)IN(04,05,11,12);
```

```
104.WQT EXTRACT INITIALS FROM FNAME AND LNAME IN BELOW FORMAT??
```

```
I/P:
```

```
FNAME:RAMYA
LNAME:AUNTY
```

```
O/P:
----
R.A.
```

```
SELECT CONCAT(SUBSTR(FNAME,1,1),'.',substr(LNAME,1,1),'.')
FROM EMPS;
```

```
105.WQTD FIRST HALF OF FNAME FROM EMPS TABLE???
```

```
SELECT SUBSTR(FNAME,1,LENGTH(FNAME)/2) AS FIRST_HALF
FROM EMPS;
```

106.WQTD 2ND HALF OF FNAME FROM EMPS TABLE??

```
SELECT SUBSTR(FNAME,LENGTH(FNAME)/2+1) AS SECOND_HALF
FROM EMPS;
```

107.WQTD 1ST HALF OF JOB IN LOWERCASE AND 2ND HALF OF JOB IN REVERSE FORMAT BASED ON SAMPLE O/P??

WAITER:waiRET

```
SELECT CONCAT(LOWER(SUBSTR(JOB,1,LENGTH(JOB)/2)),
REVERSE(SUBSTR(JOB,LENGTH(JOB)/2+1)))
FROM EMPS;
```

108.WQTD FNAME AND PASSWORD,THE PASSWORD MUST CONTAIN BELOW CONDITIONS:

```
>FIRST 2 CHARACTERS OF FNAME
>LENGTH OF FNAME
>LAST 3 CHARACTERS OF THEIR JOB
```

```
SELECT FNAME,CONCAT(SUBSTR(FNAME,1,2),LENGTH(FNAME),SUBSTR(JOB,-3,3)) AS PASSWORD
FROM EMPS;
```

109.WQTD FNAME AND PASSWORD,THE PASSWORD MUST CONTAIN BELOW CONDITIONS:

```
>2ND HALF OF STATUS
>LENGTH OF LNAME
>REVERSED LAST 3 CHARACTERS OF THEIR FNAME
```

```
SELECT FNAME,CONCAT(SUBSTR(STATUS,LENGTH(STATUS)/2+1),
LENGTH(LNAME),REVERSE(SUBSTR(FNAME,-3,3))) AS
PASSWORD
FROM EMPS;
```

```
7.REPLACE():
-------------
IT IS USED TO REPLACE SUBSTRING FROM NEW STRING IN ORIGINAL STRING.

SYNTAX:
-------
REPLACE('ORIGINAL STRING','SUBSTRING','NEW STRING');
```

```
SELECT REPLACE('PENTAGON','PENT','HEX');HEXAGON

SELECT REPLACE('PENTAGON','N','S');PESTAGOS

SELECT REPLACE('PENTAGON','R','A');PENTAGON

SELECT REPLACE('PENTAGON','N','');PETAGO

SELECT REPLACE('PENTAGON','n','R');PENTAGON
```

110.WQT REPLACE S WITH [S] IN FNAME??

SAHANA: [S]AHANA

```
SELECT REPLACE(FNAME,'S','[S]')
FROM EMPS;
```

111.WQT REPLACE A AND I WITH {A} AND {I} IN FNAME???

KIRAN:K{I}R{A}N

```
              KIR{A}N
SELECT REPLACE(REPLACE(FNAME,'A','{A}'),'I','{I}')
FROM EMPS;
```

112.WQTD COUNT OF THE CHARACTER A IN MALAYALAM???

O/P:4

```
SELECT LENGTH('MALAYALAM')-LENGTH(REPLACE('MALAYALAM','A',''));
```

113.WQTD FNAME WHICH CONTAINS EXACTLY 1 A???

```
SELECT FNAME
FROM EMPS
WHERE LENGTH(FNAME)-LENGTH(REPLACE(FNAME,'A',''))=1;
```

```
NUMBER FUNCTIONS:
-----------------
CHARACTERISTICS OF NUMBER FUNCTIONS:
------------------------------------
>IT TAKES N NUMBER OF INPUTS AND GENERATE N NUMBER OF OUTPUTS.
>IT EXECUTES ROW BY ROW
>WE CAN NEST NUMBER FUNCTIONS.
>WE CAN USE NUMBER FUNCTIONS INSIDE WHERE CLAUSE.
```

TYPES OF NUMBER FUNCTIONS:

1.ABS():

IT IS USED TO CONVERT NEGATIVE NUMBER INTO POSITIVE.

EX:

```
SELECT ABS(-18);18

SELECT ABS(18);18
```

2.MOD():

IT IS USED TO OBTAIN REMAINDER VALUE.

SYNTAX: MOD(M,D)

EX:

```
SELECT MOD(9,3);0
SELECT MOD(9,2);1
```

114.WQTD DETAILS OF EMPS WHO ARE HAVING EVEN EID???

```
SELECT *
FROM EMPS
WHERE MOD(EID,2)=0;
```

3.ROUND():

IT IS USED TO ROUND OFF THE NUMBER UPTO SPECIFIED NUMBER OF DECIMAL PLACES.

SYNTAX: ROUND(NUMBER,[DECIMAL_PLACE_VALUE]);

```
SELECT ROUND(420.4);420

SELECT ROUND(420.6);421

SELECT ROUND(420.678,2);420.68

SELECT ROUND(420.678345,4);420.6783

SELECT ROUND(420.678345,5);420.67835
```

4.CEIL:

>IT WILL OBTAIN NEXT INTEGER VALUE FROM THE GIVEN DECIMAL VALUE.
(IF IT IS POSITIVE NUMBER)
>IT WILL OBTAIN CURRENT INTEGER VALUE FROM THE GIVEN DECIMAL VALUE.
(IF IT IS NEGATIVE NUMBER)

```
SELECT CEIL(45.7);46

SELECT CEIL(45.3);46

SELECT CEIL(-45.3);-45

SELECT CEIL(-45.9);-45
```

5.FLOOR:

>IT WILL OBTAIN CURRENT INTEGER VALUE FROM GIVEN DECIMAL VALUE.
(IF IT IS POSITIVE NUMBER)
>IT WILL OBTAIN NEXT INTEGER VALUE FROM THE GIVEN DECIMAL VALUE.
(IF IT IS NEGATIVE NUMBER).

```
SELECT FLOOR(18.5);18

SELECT FLOOR(-18.5);-19
```

6.TRUNCATE():

IT IS USED CUT OFF THE VALUE UPTO SPECIFIED NUMBER OF DECIMAL PLACES WITHOUT ROUNDING IT.

SYNTAX: TRUNCATE(NUMBER,DECIMAL_PLACE_VALUE);

```
SELECT TRUNCATE(123.456,2);123.45

SELECT TRUNCATE(123.45673,4);123.4567
```

7.POW():

IT IS USED TO OBTAIN POWER VALUE OF THE NUMBER.

EX:

```
SELECT POW(2,3);8
```

8.SQRT():

IT IS USED TO OBTAIN SQUARE ROOT VALUE OF NON NEGATIVE NUMBER.

EX:

SELECT SQRT(64);8

SELECT SQRT(-64);NULL

DATE FUNCTIONS:

CHARACTERISTICS OF DATE FUNCTIONS:

>IT TAKES N NUMBER OF INPUTS AND GENERATE N NUMBER OF OUTPUTS.
>IT EXECUTES ROW BY ROW.
>WE CAN NEST DATE FUNCTIONS.
>WE CAN USE DATE FUNCTIONS INSIDE WHERE CLAUSE.

TYPES OF DATE FUNCTIONS:

1.CURDATE():

IT IS USED TO OBTAIN CURRENT DATE FROM THE SYSTEM.

EX:

SELECT CURDATE();2025-10-18

2.SYSDATE()/NOW()

IT IS USED TO OBTAIN CURRENT DATE AND TIME FROM THE SYSTEM.

EX:

SELECT SYSDATE();2025-10-18 16:10:32

SELECT NOW();2025-10-18 16:10:32

3.YEAR():

IT IS USED TO EXTRACT YEAR FROM THE GIVEN DATE EXPRESSION.

SELECT YEAR('2020-10-12');2020

115.WQTD NUMBER OF EMPS HIRED IN EACH YEAR AND ARRENGE THE NUMBERS IN MAXIMUM TO MINIMUM ORDER??

```
SELECT COUNT(*),YEAR(DOJ)
FROM EMPS
GROUP BY YEAR(DOJ)
ORDER BY COUNT(*) DESC;
```

4.MONTH():

IT IS USED TO EXTRACT MONTH VALUE FROM THE GIVEN DATE EXPRESSION.

EX: SELECT MONTH('2022-12-11');12

116.WQTD NUMBER OF EMPS BORN IN EACH MONTH IN THE YEAR 1991??

```
SELECT COUNT(*),MONTH(DOB)
FROM EMPS
WHERE YEAR(DOB)=1991
GROUP BY MONTH(DOB);
```

5.DAY():

IT IS USED TO EXTRACT DATE VALUE FROM THE GIVEN DATE EXPRESSION.

SELECT DAY('2001-12-10');10

117.WQTD DETAILS OF THE ORDERS IF ORDERS ORDERED IN THE YEAR 2023 AND IN THE MONTH EITHER DECEMBER OR OCTOBER IN THE DATE 07???

```
SELECT *
FROM ORDERS
WHERE YEAR(ORDER_DATE)=2023 AND
MONTH(ORDER_DATE)IN(12,10) AND DAY(ORDER_DATE)=07;
```

118.WQTD FIRST HIRED EMPLOYEE DOJ?

```
SELECT DOJ
FROM EMPS
ORDER BY DOJ ASC
LIMIT 1;
```

OR

```
SELECT MIN(DOJ)
FROM EMPS;
```

```
SELECT MAX(DOJ)
FROM EMPS;
```

OR

```
SELECT DOJ
FROM EMPS
ORDER BY DOJ DESC
LIMIT 1;
```

6.DATEDIFF():

IT IS USED TO OBTAIN DAY DIFFERENCE BETWEEN TWO DATE VALUES.

EX:

```
SELECT DATEDIFF('2025-10-22','2025-10-21'); 1
```

7.DATE_ADD():

IT IS USED TO ADD SOME TIME INTERVAL FOR THE GIVEN DATE VALUE.

SYNTAX:

```
DATE_ADD('DATE VALUE',INTERVAL VALUE UNIT);
```

INTERVAL:WE USE THIS KEYWORD TO ADD/SUBTRACT SOME TIME INTERVAL.

VALUE:THE AMOUNT OF TIME WHICH WE ARE ADDING.

UNIT:UNIT OF TIME INTERVAL(EX:YEAR,MONTH OR DAY)

119.WQT ADD 7 YEAR FOR THE BELOW DATE:

2020-10-12

```
SELECT DATE_aDD('2020-10-12',INTERVAL 7 YEAR);
```

120.WQT ADD 3 YEAR 6 MONTH FOR THE BELOW DATE:

2005-04-13

```
SELECT DATE_aDD('2005-04-13',INTERVAL 42 MONTH);

               2008-04-13
SELECT DATE_aDD(DATE_aDD('2005-04-13',INTERVAL 3 YEAR),INTERVAL 6 MONTH);
```

7.DATE_SUB():

IT IS USED TO SUBTRACT SOME TIME INTERVAL FROM GIVEN DATE VALUE

SYNTAX:

```
DATE_SUB('DATE VALUE',INTERVAL VALUE UNIT);
```

121.WQT SUBTRACT 2 YEAR FROM THE CURRENT DATE??

```
SELECT DATE_sUB(CURDATE(),INTERVAL 2 YEAR);
```

122.WQT SUBTRACT 5 YEAR 2 MONTH FROM THE LAST HIRED EMPLOYEE DOJ???

```
SELECT DATE_SUB(MAX(DOJ),INTERVAL 62 MONTH)
FROM EMPS;
```

8.DATE_FORMAT():

IT IS USED TO EXTRACT SOME TIME INTERVAL FROM GIVEN DATE-TIME EXPRESSION.

SYNTAX: DATE_FORMAT('DATE VALUE','DATE_FORMAT_PATTERN');

```
YYYY    :'%Y' SELECT DATE_FORMAT(NOW(),'%Y');2025
YY      :'%y' SELECT DATE_FORMAT(NOW(),'%y');25
MONTH   :'%M' SELECT DATE_FORMAT(NOW(),'%M');OCTOBER
MM      :'%m' SELECT DATE_FORMAT(NOW(),'%m');10
MON     :'%b' SELECT DATE_FORMAT(NOW(),'%b');OCT
DD      :'%d' SELECT DATE_FORMAT(NOW(),'%d');22
DAY     :'%W' SELECT DATE_FORMAT(NOW(),'%W');WEDNESDAY
DAY     :'%a' SELECT DATE_FORMAT(NOW(),'%a');WED
HH24    :'%H' SELECT DATE_FORMAT(NOW(),'%H');16
HH12    :'%h' SELECT DATE_FORMAT(NOW(),'%h');04
MINUTE  :'%i' SELECT DATE_FORMAT(NOW(),'%i');25
SECOND  :'%s' SELECT DATE_FORMAT(NOW(),'%s');08
TIME    :'%T' SELECT DATE_FORMAT(NOW(),'%T');16:25:08
AM/PM   :'%p' SELECT DATE_FORMAT(NOW(),'%p');PM
TIME
AM/PM   :'%r' SELECT DATE_FORMAT(NOW(),'%r'); 04:25:08 PM
```

```
SELECT DATE_FORMAT(NOW(),'%M %W');
```

123.WQTD CURRENT DATE AND TIME IN BELOW FORMAT:

25-OCTOBER-22 04 WED PM

```
SELECT DATE_fORMAT(NOW(),'%y-%M-%d %h %a %p');
```

124.WQTD FNAME AND DOJ IF EMPLOYEE HIRED ON SUNDAY OR FRIDAY??

```
SELECT FNAME,DOJ,DATE_FORMAT(DOJ,'%W') AS DAY_NAME
FROM EMPS
WHERE DATE_FORMAT(DOJ,'%W') IN('SUNDAY','FRIDAY');
```

RULE:

1.THE COLUMN_NAME SELECTED IN INNER QUERY AND THE COLUMN_NAME WRITTEN INSIDE OUTER QUERY SHOULD BE OF SAME DATATYPE.

```
WHERE SAL>(SELECT FNAME        : INVALID
      INT         VARCHAR(10)

WHERE SAL>(SELECT COMM         : VALID
      INT         INT
```

2.WE CAN SELECT ONLY ONE COLUMN INSIDE INNER QUERY.

```
WHERE SAL>(SELECT SAL,COMM     :INVALID
```

125.WQTD DETAILS OF THE EMPS WHO ARE GETTING SALARY LESS THAN PRIYA?

```
SELECT *
FROM EMPS
WHERE SAL<(SELECT SAL
           FROM EMPS
           WHERE FNAME='PRIYA');
```

126.WQTD FNAME,LNAME,JOB AND SALARY IF EMPS ARE WORKING SAME AS
    MURALI'S JOB ROLE??

```
SELECT FNAME,LNAME,JOB,SAL
FROM EMPS
WHERE FNAME!='MURALI' AND JOB=(SELECT JOB
                               FROM EMPS
                               WHERE FNAME='MURALI');
```

127.WQTD DETAILS OF EMPS IF EMPS FNAME 2ND CHARACTER IS A AND LNAME LAST 2ND CHARACTER IS A AND GETTING SALARY LESS THAN JAHNAVI????

```
SELECT *
FROM EMPS
WHERE FNAME LIKE '_A%' AND SUBSTR(LNAME,-2,1)='A' AND SAL<(SELECT SAL
                                                           FROM EMPS
                                                           WHERE FNAME='JAHNAVI');
      SUBSTR(FNAME,2,1)='A' AND LNAME LIKE '%A_'
```

128.WQTD DETAILS OF EMPS WHO ARE GETTING 4TH MAXIMUM SALARY??

```
SELECT *
FROM EMPS
WHERE SAL=(SELECT DISTINCT SAL
           FROM EMPS
           ORDER BY SAL DESC
           LIMIT 1 OFFSET 3);
```

129.WQTD DETAILS OF EMPS WHO ARE ELDER THAN PRIYA???

```
SELECT *
FROM EMPS
WHERE DOB<(SELECT DOB
FROM EMPS
WHERE FNAME='PRIYA');
```

1999 < 2005

130.wQTD FNAME AND LNAME TOGETHER AS FULL NAME ALONG WITH THIER SALARY,JOB AND LID IF EMPLOYEE IS WORKING IN THE LOCATION SAME AS SURESH LOCATION AND GETTING SALARY MORE THAN ARJUN???

```
SELECT CONCAT(FNAME,' ',LNAME) FULL_NAME,SAL,JOB,LID
FROM EMPS
WHERE LID=(SELECT LID
           FROM EMPS
           WHERE FNAME='SURESH') AND SAL>(SELECT SAL
                                          FROM EMPS
                                          WHERE FNAME='ARJUN');
```

131. WQTD CITY NAME OF EMPLOYEE KIRAN??

```
SELECT CITY
FROM LOCATIONS
```

```
WHERE LID=(SELECT LID
           FROM EMPS
           WHERE FNAME='KIRAN');
```

132.WQTD DETAILS OF EMPS WHO ARE LIVING IN THE STATE KARNATAKA???

```
SELECT *
FROM EMPS
WHERE LID=(SELECT LID
           FROM LOCATIONS
            WHERE STATE='KARNATAKA');
```

133.WQTD CITY AND STATE OF THE CUSTOMER VIRAT KOHLI???

```
SELECT CITY,STATE
FROM LOCATIONS
WHERE LID=(SELECT LID
           FROM CUSTOMERS
           WHERE NAME='VIRAT KOHLI');
```

134.WQTD ITEM DETAILS WHICH BELONGS TO SPICE HUB RESTAURANT???

```
SELECT *
FROM MENU_ITEMS
WHERE RESTAURANT_ID=(SELECT RESTAURANT_ID
                     FROM RESTAURANTS
                     WHERE NAME='SPICE HUB');
```

135.WQTD RATING OF THE CUSTOMER WHO HAS GIVEN FROM KOLKATA CITY??

```
SELECT RATING
FROM REVIEWS
WHERE CID=(SELECT CID
           FROM CUSTOMERS
           WHERE LID =(SELECT LID
                       FROM LOCATIONS
                       WHERE CITY='KOLKATA'));
```

136.WQTD DETAILS OF CUSTOMERS WHO ARE LIVING IN CHENNAI OR JAIPUR CITY?

```
SELECT *
FROM CUSTOMERS
WHERE LID IN(SELECT LID
           FROM LOCATIONS
           WHERE CITY IN('CHENNAI','JAIPUR'));
```

```
TYPES OF SUBQUERY:
-------------------
1.SINGLE-ROW SUBQUERY
2.MULTI-ROW SUBQUERY
3.CO-RELATED SUBQUERY
```

```
1.SINGLE ROW SUBQUERY:
----------------------
IF INNER QUERY RETURNS SINGLE VALUE THEN WE CAN CONSIDER IT AS SINGLE ROW SUBQUERY.
```

```
>IT SUPPORTS BOTH NORMAL OPERATORS(=,!=,<,>) AND
SPECIAL OPERATORS(IN,NOT IN,ALL,ANY)
```

```
2.MULTI ROW SUBQUERY:
---------------------
IF INNER QUERY RETURNS MORE THAN ONE VALUE THEN WE CAN CONSIDER IT AS MULTI ROW SUBQUERY.
```

```
>IT SUPPORTS ONLY SPECIAL OPERATORS(IN,NOT IN,ALL,ANY)
```

137.WQTD DETAILS OF CUSTOMERS WHO ORDERED SOME ITEM???

```
SELECT *
FROM CUSTOMERS
WHERE ORDER_ID IS NOT NULL;
```

OR

```
SELECT *
FROM CUSTOMERS
WHERE ORDER_ID IN(SELECT ORDER_ID
                  FROM ORDERS);
```

138.WQTD DETAILS OF THE EMPS WHO ARE GETTING SALARY LESS THAN ALL THE WAITERS?

```
SELECT *
FROM EMPS
WHERE SAL <ALL(SELECT SAL
           FROM EMPS
           WHERE JOB='WAITER');
```

ALL:

IT IS A MULTIVALUE OPERATOR WHICH TAKES MULTIPLE VALUES AT THE RHS AND SINGLE VALUE AT THE LHS ALONG WITH RELATIONAL OPERATORS.

SYNTAX:

```
          LHS                                    RHS
COLUMN_NAME/EXPRESSION < / > / <= / >= ALL(V1,V2,......VN);
```

>IT WORKS ON AND OPERATOR.

A:1000   T
B:2000   T
C:10000 F

D:WAITER:2000
E:WAITER:4000
F:WAITER:10000

SAL<ANY(2000,4000,10000)

1000<ANY(
2000<ANY(
10000<ANY(

139.WQTD DETAILS OF EMPS WHO ARE GETTING SALARY LESS THAN ANY ONE OF THE WAITER??

ANY:

IT IS A MULTI VALUE OPERATOR WHICH TAKES MULTIPLE VALUES AT THE RHS AND SINGLE VALUE AT THE LHS ALONG WITH RELATIONAL OPERATORS.

SYNTAX:

```
COLUMN_NAME/EXPRESSION < / > / <= / >= ANY(V1,V2,......VN);
```

>IT WORKS ON OR OPERATOR.

```
SELECT *
FROM EMPS
WHERE SAL <ANY(SELECT SAL
           FROM EMPS
           WHERE JOB='WAITER');
```

140.WQTD DETAILS OF EMPS WHO DELIVERED THE ITEM TO THE CUSTOMER WHO BELONGS TO KOLKATA CITY???

```
SELECT *
FROM EMPS
WHERE EID IN(SELECT EID
             FROM ORDERS
WHERE STATUS='DELIVERED' AND ORDER_ID IN(SELECT ORDER_ID
                                         FROM CUSTOMERS
                                         WHERE LID IN(SELECT LID
                                                      FROM LOCATIONS
                                                      WHERE CITY='KOLKATA')));
```

141.WQTD DETAILS OF MURALI'S MANAGER??

```
SELECT *
FROM EMPS
WHERE EID IN(SELECT MGR
             FROM EMPS
             WHERE FNAME='MURALI');
```

142.WQTD CITY NAME OF DIVYA'S MANAGER???

```
SELECT CITY
FROM LOCATIONS
WHERE LID IN(SELECT LID
             FROM EMPS
             WHERE EID IN(SELECT MGR
                          FROM EMPS
                          WHERE FNAME='DIVYA'));
```

143.WQTD DETAILS OF AMAN'S MANAGER'S MANAGER???

```
SELECT *
FROM EMPS
WHERE EID IN(SELECT MGR
             FROM EMPS
             WHERE EID IN(SELECT MGR
                          FROM EMPS
                          WHERE FNAME='AMAN'));
```

144.WQTD DETAILS OF EMPS WHO ARE REPORTING TO JAHNAVI???

```
SELECT *
FROM EMPS
WHERE MGR IN(SELECT EID
             FROM EMPS
             WHERE FNAME='JAHNAVI');
```

145.WQTD DETAILS OF EMPS WHO ARE REPORTING TO ARJUN'S MANAGER???

```
SELECT *
FROM EMPS
WHERE MGR IN(SELECT EID
             FROM EMPS
             WHERE EID IN(SELECT MGR
                          FROM EMPS
                          WHERE FNAME='ARJUN'));
```

146.WQTD LOCATION DETAILS OF THE EMPS WHO ARE REPORTING TO FAIZAN'S MANAGER'S MANAGER??

```
SELECT *
FROM LOCATIONS
WHERE LID IN(SELECT LID
             FROM EMPS
             WHERE MGR IN(SELECT EID
                          FROM EMPS
                          WHERE EID IN(SELECT MGR
                                       FROM EMPS
                                       WHERE EID IN(SELECT MGR
                                                    FROM EMPS
                                                    WHERE FNAME='FAIZAN'))));
```

```
DRAWBACK OF SUBQUERY:
---------------------
>WE CANT RETRIEVE THE DATA FROM MULTIPLE TABLES SIMULTANEOUSELY.

JOINS:
------
IT IS USED TO RETRIEVE THE DATA FROM MULTIPLE TABLES SIMULTANEOUSELY.

TYPES OF JOINS:
---------------
1.CROSS JOIN/CARTESIAN JOIN
2.INNER JOIN
3.OUTER JOIN
            i.LEFT OUTER JOIN/LEFT JOIN
            ii.RIGHT OUTER JOIN/RIGHT JOIN
4.SELF JOIN
5.NATURAL JOIN

1.CROSS JOIN:
-------------
IT IS USED TO MERGE THE RECORDS OF ONE TABLE WITH RECORDS OF ANOTHER TABLE.

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SYNTAX:

SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME T1 CROSS JOIN TABLE_NAME T2;
```

147.WQTD CARTESIAN PRODUCT FROM LOCATIONS AND EMPS TABLE????

```
SELECT *
FROM LOCATIONS L CROSS JOIN EMPS E;

SELECT L.LID,E.LID
FROM LOCATIONS L CROSS JOIN EMPS E;

DRAWBACK:
---------
INNER JOIN:
-----------
IT IS USED TO OBTAIN MATCHED RECORDS FROM DIFFERENT TABLES.

SYNTAX:
-------
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME T1 INNER JOIN TABLE_NAME T2
ON JOIN_CONDITION;

ON:IT IS USED TO PASS JOIN_CONDITION.

JOIN_CONDITION:IT IS USED TO CONNECT DIFFERENT TABLES.
```

```
T1.COLUMN_NAME=T2.COLUMN_NAME
```

148.WQTD EMPLOYEE FNAME ALONG WITH THIER CITY NAME ???

```
EMPS:E
LOCATIONS:L

SELECT E.FNAME,L.CITY
FROM EMPS E INNER JOIN LOCATIONS L
ON E.LID=L.LID;
```

149.WQTD ITEM NAME AND RESTAURANT NAME IF ITEM NAME IS FISH CURRY???

```
MENU_ITEMS:M
RESTAURANTS:R

SELECT M.NAME ITEM_NAME,R.NAME RESTAURANT_NAME
FROM MENU_ITEMS M INNER JOIN RESTAURANTS R
ON M.RESTAURANT_ID=R.RESTAURANT_ID
WHERE M.NAME='FISH CURRY';
```

150.WQTD AVERAGE RATING PER RESTAURANT NAME???

```
REVIEWS:R
RESTAURANTS:RR

SELECT AVG(R.RATING),RR.NAME
FROM REVIEWS R INNER JOIN RESTAURANTS RR
ON R.RESTAURANT_ID=RR.RESTAURANT_ID
GROUP BY RR.NAME;
```

151.WQTD RESTAURANTS WHICH GOT HIGHEST RATING???

```
RESTAURANTS:R
REVIEWS:RR

SELECT R.NAME,RR.RATING
FROM RESTAURANTS R INNER JOIN REVIEWS RR
ON R.RESTAURANT_ID=RR.RESTAURANT_ID
WHERE RR.RATING=(SELECT MAX(RATING)
                FROM REVIEWS);
```

152.WQTD CUSTOMER NAME AND THIER ORDER DETAILS AND DELIVERY PERSON NAME?

```
CUSTOMERS:C
ORDERS:O
EMPS:E

SELECT C.NAME CUSTOMER_NAME,O.*,E.FNAME DELIVERY_PERSON_NAME
FROM CUSTOMERS C INNER JOIN ORDERS O INNER JOIN EMPS E
ON C.ORDER_ID=O.ORDER_ID AND O.EID=E.EID;
```

153.WQTD CUSTOMER NAME AND THIER ORDER DETAILS WITH PENDING PAYMENTS AND ACCEPTED STATUS??

```
CUSTOMERS:C
ORDERS:O
PAYMENTS:P

SELECT C.NAME,O.*,P.STATUS PAYMENT_STATUS
FROM CUSTOMERS C INNER JOIN ORDERS O INNER JOIN PAYMENTS P
ON C.ORDER_ID=O.ORDER_ID AND O.ORDER_ID=P.ORDER_ID
WHERE P.STATUS='PENDING' AND O.STATUS='ACCEPTED';
```

154.WQTD EMPS WHO DELIVERED MOST ORDERS??

```
SELECT E.FNAME,COUNT(*),E.EID
FROM EMPS E INNER JOIN ORDERS O
ON E.EID=O.EID
WHERE O.STATUS='DELIVERED'
GROUP BY E.FNAME,E.EID
HAVING COUNT(*)=(SELECT COUNT(*)
                 FROM ORDERS
                 WHERE STATUS='DELIVERED'
                 GROUP BY EID
                 ORDER BY COUNT(*) DESC
                 LIMIT 1);
```

155.WQTD CUSTOMERS WHO MADE THE HIGHEST PAYMENT???

```
PAYMENTS:P
CUSTOMERS:C
ORDERS:O

SELECT SUM(P.AMOUNT),C.NAME,C.CID
FROM CUSTOMERS C INNER JOIN PAYMENTS P INNER JOIN ORDERS O
ON C.ORDER_ID=O.ORDER_ID AND O.ORDER_ID=P.ORDER_ID
```

```
WHERE P.STATUS='COMPLETED'
GROUP BY C.NAME,C.CID
HAVING SUM(P.AMOUNT)=(SELECT SUM(P.AMOUNT)
                      FROM CUSTOMERS C INNER JOIN PAYMENTS P INNER JOIN ORDERS O
                      ON C.ORDER_ID=O.ORDER_ID AND O.ORDER_ID=P.ORDER_ID
                      WHERE P.STATUS='COMPLETED'
                      GROUP BY C.NAME
                      ORDER BY SUM(P.AMOUNT) DESC
                      LIMIT 1);
```

```
OUTER JOIN
----------
1.LEFT OUTER JOIN
2.RIGHT OUTER JOIN
```

```
1.LEFT OUTER JOIN:
------------------
IT IS USED TO OBTAIN MATCHED AND UNMATCHED RECORDS OF LEFT TABLE.
```

```
SYNTAX:
```

```
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME T1 LEFT OUTER JOIN TABLE_NAME T2
ON T1.COLUMN_NAME=T2.COLUMN_NAME;
```

```
customers:LEFT
locations:RIGHT
```

```
156.WQTD MATCHED AND UNMATCHED RECORDS OF CUSTOMERS TABLE??
```

```
SELECT *
FROM CUSTOMERS C LEFT OUTER JOIN LOCATIONS L
ON C.LID=L.LID;
```

```
2.RIGHT OUTER JOIN
------------------
IT IS USED TO OBTAIN MATCHED AND UNMTACHED RECORDS OF RIGHT TABLE.
```

```
SYNTAX:
```

```
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME T1 RIGHT OUTER JOIN TABLE_NAME T2
ON T1.COLUMN_NAME=T2.COLUMN_NAME;
```

```
CUSTOMERS:LEFT
LOCATIONS:RIGHT
```

```
157.WQTD MATCHED AND UNMATCHED RECORDS OF LOCATIONS TABLE???
```

```
SELECT *
FROM CUSTOMERS C RIGHT OUTER JOIN LOCATIONS L
ON C.LID=L.LID;
```

```
CASE:2
-------
WE HAVE TO GIVE THE PRIORITY FOR THE TABLE WHERE COMMEN COLUMN IS PRIMARY KEY
```

```
158.WQTD CITY NAME WHERE NO CUSTOMERS ARE LIVING??
```

```
SELECT L.CITY
FROM LOCATIONS L LEFT OUTER JOIN CUSTOMERS C
ON L.LID=C.LID
WHERE C.LID IS NULL;
```

```
159.WQTD CUSTOMER NAME WHO HAS NOT GIVEN REVIEW???
```

```
SELECT C.NAME
FROM REVIEWS R RIGHT OUTER JOIN CUSTOMERS C
ON R.CID=C.CID
WHERE R.CID IS NULL;
```

```
160.WQTD RESTAURANT NAME WHICH DONT HAVE ANY REVIEW???
```

```
SELECT R.NAME
FROM RESTAURANTS R LEFT OUTER JOIN REVIEWS RR
ON R.RESTAURANT_ID=RR.RESTAURANT_ID
WHERE RR.CID IS NULL;
```

```
SELF JOIN
---------
IT IS USED TO OBTAIN MATCHED RECORDS FROM SAME TABLES.
```

```
SYNTAX:
-------
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME T1 JOIN TABLE_NAME T2
```

```
ON JOIN_CONDITION;
```

161.WQTD EMPS FNAME AND THEIR MANAGERS FNAME FROM EMPS TABLE???

```
E1:FOR EMPS
E2:FOR MANAGERS
```

```
SELECT E1.FNAME EMP_NAME,E2.FNAME MANAGER_NAME
FROM EMPS E1 JOIN EMPS E2
ON E1.MGR=E2.EID;
```

162.WQTD EMPLOYEE FNAME,SALARY AND THIER MANAGER FNAME,SALARY IF EMPLOYEE IS GETTING SALARY MORE THAN 35000 AND MANAGER IS GETTING SALARY LESS THAN 100000??

```
E1:FOR EMPS
E2:FOR MANAGERS
```

```
SELECT E1.FNAME EMP_NAME,E1.SAL EMP_SALARY,E2.FNAME MANAGER_NAME,E2.SAL MANAGER_SALARY
FROM EMPS E1 JOIN EMPS E2
ON E1.MGR=E2.EID
WHERE E1.SAL>35000 AND E2.SAL<100000;
```

163.WQTD EMPLOYEE FNAME,DOB AND MANAGER FNAME,DOB IF EMPLOYEE IS ELDER THAN THEIR MANAGER??

```
E1:FOR MANAGERS
E2:FOR EMPS
```

```
SELECT E2.FNAME,E2.DOB,E1.FNAME,E1.DOB
FROM EMPS E1 JOIN EMPS E2
ON E2.MGR=E1.EID
WHERE E2.DOB<E1.DOB;
```

164.WQTD EMPLOYEE FNAME AND EMPLOYEE BIRTH YEAR,MANAGER FNAME AND MANAGER BIRTH YEAR IF EMPLOYEE AND HIS MANAGER BORN IN THE SAME YEAR???

```
E1:FOR EMPS
E2:FOR MANAGERS
```

```
SELECT E1.FNAME,YEAR(E1.DOB) EMP_BIRTH_YEAR,E2.FNAME,YEAR(E2.DOB) MANAGER_BIRTH_YEAR
FROM EMPS E1 JOIN EMPS E2
ON E1.MGR=E2.EID
WHERE YEAR(E1.DOB)=YEAR(E2.DOB) ;
```

165.WQTD FNAME OF AMAN'S MANAGER'S MANAGER???

```
E1:FOR AMAN
E2:FOR AMAN MANAGER(A)
E3:FOR AMAN MANAGER MANAGER(B)
```

```
SELECT E3.FNAME
FROM EMPS E1 JOIN EMPS E2 JOIN EMPS E3
ON E1.MGR=E2.EID AND E2.MGR=E3.EID
WHERE E1.FNAME='AMAN';
```

166.WQTD DETAILS OF EMPS WHO ARE GETTING SALARY MORE THAN FAIZAN???

```
E1:FOR EMPS
E2:FOR FAIZAN
```

```
SELECT E1.*
FROM EMPS E1 JOIN EMPS E2
ON E1.SAL>E2.SAL
WHERE E2.FNAME='FAIZAN';
```

167.WQTD DETAILS OF EMPS WHO ARE REPORTING TO PRIYA IF PRIYA IS GETTING SALARY MORE THAN KIRAN???

```
E1:FOR EMPS
E2:FOR PRIYA
E3:FOR KIRAN
```

```
SELECT E1.*
FROM EMPS E1 JOIN EMPS E2 JOIN EMPS E3
ON E1.MGR=E2.EID AND E2.SAL>E3.SAL
WHERE E2.FNAME='PRIYA' AND E3.FNAME='KIRAN';
```

```
NATURAL JOIN:
-------------
IT IS USED TO PRINT MATCHED RECORDS FROM MULTIPLE TABLES BASED ON COMMEN ATTRIBUTE.
```

SYNTAX:

```
SELECT COLUMN_NAME/EXPRESSION
FROM TABLE_NAME T1 NATURAL JOIN TABLE_NAME T2;
```

```
SELECT *
FROM EMPS E NATURAL JOIN LOCATIONS L;
```

```
SELECT *
```

```
FROM EMPS E NATURAL JOIN EMPS L;
```

```
SET OPERATORS:
--------------
1.UNION
2.UNION ALL
3.INTERSECTION
```

```
A={1,2,3,4}
B={4,5,6}
```

```
A UNION B={1,2,3,4,5,6}
```

```
A UNION ALL B={1,2,3,4,4,5,6}
```

```
A INTERSECTION B={4}
```

```
1.UNION:
--------
IT IS USED TO RETRIVE THE RECORDS FROM MULTIPLE TABLES VERTICALLY.
```

```
>IT WILL AVOID DUPLICATE RECORDS FROM THE O/P.
```

```
2.UNION ALL
------------
IT IS USED TO RETRIVE THE RECORDS FROM MULTIPLE TABLES VERTICALLY.
```

```
>IT WILL INLCUDE THE DUPLICATE RECORDS IN THE O/P.
```

```
(SELECT FNAME
FROM EMPS)
UNION
(SELECT FNAME
FROM EMPS);    O/P: 10 FNAME
```

```
(SELECT FNAME
FROM EMPS)
UNION ALL
(SELECT FNAME
FROM EMPS);    O/P: 20 FNAME
```

```
RULES:
------
>WE SHOULD USE SEMI COLEN FOR THE LAST QUERY.
>WE SHOULD USE ROUND BRACES FOR ALL THE QUERIES(SEPERATELY).
>WE SHOULD USE SAME NUMBER OF COLUMNS IN SELECT CLAUSES.
```

```
168.WQTD DETAILS OF 3RD AND 5TH RECORD FROM EMPS TABLE???
```

```
(SELECT *
FROM EMPS
LIMIT 1 OFFSET 2)
UNION
(SELECT *
FROM EMPS
LIMIT 1 OFFSET 4);
```

```
169.WQTD DETAILS OF 4TH,5TH,7TH AND 10TH RECORD FROM EMPS TABLE??
```

```
(SELECT *
FROM EMPS
LIMIT 2 OFFSET 3)
UNION
(SELECT *
FROM EMPS
LIMIT 1 OFFSET 6)
UNION
(SELECT *
FROM EMPS
LIMIT 1 OFFSET 9);
```

```
170.WQTD MATCHED AND UNMATCHED RECORDS FROM BOTH CUSTOMERS AND LOCATIONS?
```

```
CUSTOMERS:LEFT
LOCATIONS:RIGHT
```

```
(SELECT *
FROM CUSTOMERS C LEFT OUTER JOIN LOCATIONS L
ON C.LID=L.LID)
UNION
(SELECT *
FROM CUSTOMERS C RIGHT OUTER JOIN LOCATIONS L
ON C.LID=L.LID);
```

```
CREATE DATABASE DEMO;
```

TO DISABLE AUTOCOMMIT:

SET AUTOCOMMIT=0;

```
CREATE TABLE STUDENT
(ID INT,
 NAME VARCHAR(20)
);
```

```
INSERT INTO STUDENT VALUES(1,'YASH'),(2,'DARSHAN');
```

BASICALLY,ALL THE DDL COMMANDS ARE AUTOCOMMIT COMMANDS.

TCL(TRANSACTION CONTROL LANGUAGE)

1.COMMIT
2.ROLLBACK
3.SAVEPOINT

1.COMMIT:

IT IS USED TO SAVE TRANSACTIONS(DML OPERATIONS) PERMENENTLY INSIDE THE DATABASE.

SYNTAX:

```
COMMIT;
```

```
UPDATE STUDENT
SET NAME='D TOSS'
WHERE ID=2;
```

2.ROLLBACK:

IT IS USED TO ROLL OUT THE OPERATIONS UPTO PREVIOUSELY USED COMMIT STATEMENT.

SYNTAX:

```
ROLLBACK;
```

>WE CAN GETBACK DELETED RECORDS BY USING ROLLBACK IF COMMIT IS NOT USED AFTER DELETE OPERATION.

```
INSERT INTO STUDENT VALUES(5,'ALLU ARJUN');
```

>WE CANT ROLL OUT THE OPERATIONS UPTO CERTAIN COMMIT STATEMENTS.

>WE CAN ROLL OUT THE OPERATIONS UPTO CERTAIN SAVEPOINTS.

SYNTAX:

```
ROLLBACK TO SAVEPOINT_NAME;
```

3.SAVEPOINT:

>IT IS USED TO CREATE SAVEPOINTS IN BETWEEN THE TRANSACTIONS.

SYNTAX:

```
SAVEPOINT SAVEPOINT_NAME;
```

>IT WILL NOT SAVE THE TRANSACTIONS PERMENENTLY INSIDE THE DATABASE.

DCL(DATA CONTROL LANGUAGE)

1.GRANT
2.REVOKE

1.GRANT:

IT IS USED GRANT/PROVIDE THE PERMISSION OF THE DATA FROM ONE USER TO ANOTHER USER.

SYNTAX:

```
GRANT SQL_STATEMENTS ON TABLE_NAME TO 'USERNAME'@'HOSTNAME';
```

TO CHECK ALL THE USERS PRESENT IN MYSQL

STEP:1

```
USE INFORMATION_SCHEMA;
```

STEP:2

```
SELECT *
FROM USER_ATRIBUTES;
```

TO CHECK ACTIVE USER PRESENT IN MYSQL:

```
SELECT USER();

TO CREATE USER IN MYSQL:
------------------------
SYNTAX:
-------

CREATE USER 'USERNAME'@'HOSTNAME'IDENTIFIED BY'PASSWORD';

HOSTNAME:LOCALHOST,%

USERNAME:PENTAGON
HOSTNAME:LOCALHOST
PASSWORD:SQL

CREATE USER 'PENTAGON'@'LOCALHOST'IDENTIFIED BY'SQL';

TO ACCESS MYSQL ACCOUNT IN COMMAND PROMPT:
------------------------------------------
SYNTAX:

mysql -u username -p

GRANT SELECT ON EMPS TO 'PENTAGON'@'LOCALHOST';

GRANT UPDATE ON EMPS TO 'PENTAGON'@'LOCALHOST';

ALL:IT IS USED GRANT ALL THE PERMISSIONS AT A TIME

GRANT ALL ON EMPS TO 'PENTAGON'@'LOCALHOST';

REVOKE:
-------
IT IS USED TO GETBACK PERMISSION OF THE DATA FROM ANOTHER USER.

SYNTAX:
-------
REVOKE SQL_STATEMENTS ON TABLE_NAME FROM 'USERNAME'@'HOSTNAME';

REVOKE DELETE ON EMPS FROM 'PENTAGON'@'LOCALHOST';

REVOKE ALL ON EMPS FROM 'PENTAGON'@'LOCALHOST';

TO DROP USER FROM MYSQL:
------------------------
SYNTAX:

DROP USER 'USERNAME'@'HOSTNAME';

1.is it POSSIBLE TO CREATE DUPLICATE TABLE???

YES.....

SYNTAX:
-------
CREATE TABLE NEW_TABLE_NAME(SELECT * FROM TABLE_NAME);

EMPS1:EMPS

CREATE TABLE EMPS1(SELECT * FROM EMPS);

2.IS IT POSSIBLE TO CREATE DUPLICATE TABLE WITHOUT RECORDS????

YES....

SYNTAX:
-------
CREATE TABLE NEW_TABLE_NAME(SELECT * FROM EMPS WHERE FALSE_CONDITION);

EMPS2:EMPS

CREATE TABLE EMPS2(SELECT * FROM EMPS WHERE 1=0);

3.IS IT POSSIBLE TO ADD RECORDS FROM ONE TABLE TO ANOTHER TABLE???
YESS....(TABLES STRUCTURE SHOULD BE SAME)

SYNTAX:
-------

INSERT INTO TABLE_NAME(SELECT STATEMENT);

EMPS2:DELIVERY BOYS DATA FROM EMPS TABLE

INSERT INTO EMPS2(SELECT * FROM EMPS WHERE JOB='DELIVERY');

INSERT INTO EMPS2(SELECT * FROM LOCATIONS WHERE CITY='MUMBAI');ERROR

SUBTABLE:
---------
WAITERS_DATA(WAITERS DATA FROM EMPS TABLE)

CREATE TABLE WAITERS_DATA(SELECT * FROM EMPS WHERE JOB='WAITER');
```

VIEW:

>IT IS A VIRTUAL TABLE.
>IT DOES'NT OCCUPY ANY MEMORY INSIDE THE DATABASE.
>TO OVERCOME THE DRAWBACK OF SUBTABLE WE USE VIEW.

TO CREATE VIEW:

SYNTAX:

```
CREATE VIEW VIEWNAME AS(SELECT STATMENT);
```

WAITER:WAITERS DATA FROM EMPS TABLE

```
CREATE VIEW WAITER AS(SELECT * FROM EMPS WHERE JOB='WAITER');
```

TO DROP VIEW:

SYNTAX:

```
DROP VIEW VIEW_NAME;
```

```
DROP VIEW WAITER;
```

CASE STATEMENT:

IT IS USED TO PASS CERTAIN CONDITIONS WHILE DISPLAYING THE RECORDS.

SYNTAX:

```
SELECT CASE
            WHEN CONDITION_1 THEN VALUE_1
            WHEN CONDITION_2 THEN VALUE_2
             '
             '
            WHEN CONDITION_N THEN VALUE_N

       ELSE DEFAULT_VALUE
       END AS ALIAS_NAME

FROM TABLE_NAME;
```

171.WQTD FNAME AND IF GENDER='M' PRINT IT AS SIGMA_MALE AND IF GENDER='F' PRINT IT AS DADS LIL PRINCESS
ELSE PRINT IT AS OTHER.

ROLE:

```
SELECT FNAME,CASE
                WHEN GENDER='M' THEN 'SIGMA_MALE'
                WHEN GENDER='F' THEN 'DADS LIL PRINCESS'
             ELSE 'OTHER'
             END AS ROLE
FROM EMPS;
```

172.WQTD FNAME,DOJ AND IF EMPS HIRED BEFORE 2020 PRINT IT AS SENIOR ELSE PRINT IT AS JUNIOR??

ROLE:

```
SELECT FNAME,DOJ,CASE
                    WHEN DOJ<'2020-01-01' THEN 'SENIOR'
                ELSE 'JUNIOR'
                END AS ROLE
FROM EMPS;
```

```
YEAR(DOJ)<2020
```

173.WQTD FNAME OF EMPS WHO ARE GETTING SALARY MORE THAN AVERAGE SALARY IN
    THIER JOB ROLE??

E1:FOR EMPS
E2:TO FIND AVG SALARY

```
SELECT E1.FNAME
FROM EMPS E1
WHERE E1.SAL>(SELECT AVG(E2.SAL)
              FROM EMPS E2
              WHERE E1.JOB=E2.JOB);
```

CO-RELATED SUBQUERY:

>BOTH OUTER QUERY AND INNER QUERY MUTUALLY DEPENDS ON EACH OTHER.

WORKING PRINCIPLE OF CO-RELATED SUBQUERY:

>OUTER QUERY EXECUTES FIRST PARTIALLY.

>INNER QUERY WILL EXECUTES FOR EACH ROW OF OUTER QUERY TABLEE1.
>OUTER QUERY EXECUTES COMPLETELY RETURN'S FINAL O/P

174.WQTD FNAME OF EMPS WHO ARE YOUNGER THAN THEIR MANAGERS???

E1:FOR EMPS
E2:FOR MANAGERS

```
SELECT E1.FNAME
FROM EMPS E1
WHERE E1.DOB>(SELECT E2.DOB
              FROM EMPS E2
              WHERE E1.MGR=E2.EID);
```

RANKING FUNCTIONS/WINDOW FUNCTIONS

IT IS USED TO ASSIGN RANKS FOR ALL THE RECORDS IN A TABLE.

SYNTAX:

```
SELECT RANKING_FUNCTION() OVER([PARTITION BY COLUMN_NAME] ORDER BY COLUMN_NAME ASC/DESC)
FROM TABLE_NAME;
```

OVER:IT IS USED TO PASS RANKING FUNCTIONS IN SELECT CLAUSE.

PARTITION BY:IT IS USED TO CREATE GROUPS,IT RESETS THE RANK AFTER EACH GROUP.

TYPES OF RANKING FUNCTIONS:

1.ROW_NUMBER()
2.RANK()
3.DENSE_RANK()

1.ROW_NUMBER()

IT IS USED TO ASSIGN UNIQUE RANKS FOR ALL THE RECORDS IN A TABLE.

SYNTAX:

```
SELECT ROW_NUMBER() OVER(PARTITION BY COLUMN_NAME ORDER BY COLUMN_NAME ASC/DESC)
FROM TABLE_NAME;
```

```
SELECT FNAME,JOB,SAL,ROW_NUMBER() OVER(PARTITION BY JOB ORDER BY SAL DESC) RANKS
FROM EMPS;
```

```
SELECT FNAME,JOB,SAL,ROW_NUMBER() OVER( ORDER BY SAL DESC) RANKS
FROM EMPS;
```

DRAWBACK:

IT WILL ASSIGN DIFFERENT RANKS FOR TIED RECORDS.

2.RANK()

IT IS USED TO ASSIGN RANKS FOR ALL THE RECORDS IN A TABLE

>IT WILL ASSIGN SAME RANK FOR TIED RECORDS.

SYNTAX:

```
SELECT RANK() OVER(PARTITION BY COLUMN_NAME ORDER BY COLUMN_NAME ASC/DESC)
FROM TABLE_NAME;
```

```
SELECT FNAME,JOB,SAL,RANK() OVER( ORDER BY SAL DESC) RANKS
FROM EMPS;
```

DRAWBACK:

IT SKIPS NEXT RANKING NUMBERS.

3.DENSE_RANK():

IT IS USED TO ASSIGN RANKS FOR ALL THE RECORDS IN A TABLE.

>IT WILL ASSIGN SAME RANK FOR TIED RECORDS ALSO IT REMAINS NEXT RANKING NUMBER IN SEQUENTIAL ORDER.

SYNTAX:

```
SELECT DENSE_RANK() OVER(PARTITION BY COLUMN_NAME ORDER BY COLUMN_NAME ASC/DESC)
FROM TABLE_NAME;
```

```
SELECT FNAME,JOB,SAL,DENSE_RANK() OVER( ORDER BY SAL DESC) RANKS
FROM EMPS;
```